

Bose–Einstein condensation at positive temperature in the dilute hard-sphere gas

OpenAI

October 5, 2026

Abstract

We prove Bose–Einstein condensation at positive temperature in the three-dimensional dilute hard-sphere gas. For each fixed exclusion distance and every sufficiently small fixed density, there is a strictly positive temperature, independent of the volume, at which the exact canonical Gibbs state has a positive condensate fraction in the thermodynamic limit. The condensate occupies the constant orbital.

Contents

1	Introduction	2
1.1	The model and the result	2
1.2	Historical context and related methods	3
1.3	The argument	4
2	A scaled formulation	5
2.1	Positive kernels and retained particles	6
2.2	The estimate to be proved	7
3	Local kinetic energy and particle-number costs	7
3.1	Two local consequences of the hard core	7
3.2	Entropy and admissible marginals	9
3.3	Insertion without changing the old marginal	9
4	Localization and joint spatial estimates	12
4.1	Localization and diagonal tilting	12
4.2	A local penalty and density control in a field	13
4.3	Simultaneous shortages are exponentially unlikely	15
4.4	Simultaneous failures of separation	16
5	A thermal slab and joint trajectory estimates	17
5.1	The path law and its purification	18
5.2	Regular trajectories and certificates	19
5.3	A localized kernel for one marked path	20
5.4	Colors and the trace-sandwich estimate	22
5.5	The interacting trajectory moment	25
5.6	Completion of the certificate estimate	27

6	Moving a path through a neighboring cell	28
6.1	Separated endpoints and Gaussian densities	29
6.2	Collision with a deterministic obstacle	29
6.3	Many paths can be moved successfully	32
7	Exposing the environment and retaining movable groups	32
7.1	Eligible labels and their local trial laws	33
7.2	Disjoint groups and the information retained	33
7.3	Accessibility and the probability of bad cells	35
7.4	One-path comparisons at fixed exposed data	35
8	Paths through a sparse set of obstacles	37
8.1	A folded random skeleton	37
8.2	Finite components and their exterior boundaries	40
8.3	Nearest-neighbor detours	41
8.4	Averaging the environment and applying both moments	42
9	A common measure and positive occupation	44
9.1	Affinity and a common finite measure	44
9.2	Cell measures and disjoint groups of tags	45
9.3	Two midpoint assignments with the same observed bath	46
9.4	Cancellation of conditional normalizers	47
9.5	The likelihood cost is confined to encounters	50
9.6	Averaging labels and forgetting the paths	51
10	Return to fixed physical parameters	53

1 Introduction

Bose–Einstein condensation is a statement about coherence across a macroscopic sample. For an interacting gas, local density estimates or accurate thermodynamic energies do not by themselves give that coherence. The question addressed here is whether a dilute three-dimensional gas with an impenetrable core has a macroscopically occupied orbital at a positive temperature that stays fixed as the volume grows.

1.1 The model and the result

Fix an exclusion distance $a > 0$. On the flat torus $\Lambda_L = (\mathbb{R}/L\mathbb{Z})^3$, let d_L denote Euclidean torus distance and set

$$\Omega_{N,L,a} = \{(x_1, \dots, x_N) \in \Lambda_L^N : d_L(x_i, x_j) > a \text{ whenever } i \neq j\}.$$

In units $\hbar^2/(2m) = k_B = 1$, the hard-sphere Hamiltonian $H_{N,L,a}$ is the self-adjoint operator associated with

$$q[f] = \sum_{i=1}^N \int_{\Omega_{N,L,a}} |\nabla_i f|^2, \quad f \in H_0^1(\Omega_{N,L,a}) \text{ symmetric.}$$

For $T > 0$, its canonical state and one-particle density matrix are

$$\Gamma_{N,L,T} = \frac{e^{-H_{N,L,a}/T}}{\text{Tr } e^{-H_{N,L,a}/T}}, \quad \gamma_{N,L,T}^{(1)} = N \text{Tr}_{2,\dots,N} \Gamma_{N,L,T}.$$

We use zero extension from the allowed configuration set when taking partial traces. Thus $\gamma_{N,L,T}^{(1)}$ acts on $L^2(\Lambda_L)$ and has trace N . The constant orbital is $u_{0,L} = L^{-3/2}$. Macroscopic occupation of this orbital means that $N^{-1}\langle u_{0,L}, \gamma_{N,L,T}^{(1)} u_{0,L} \rangle$ has a positive lower limit as $L \rightarrow \infty$ with N/L^3 tending to a fixed density. Translation invariance makes the constant orbital an eigenvector of the density matrix, so this gives condensation in the sense of Penrose and Onsager [37].

Theorem 1.1. *For every $a > 0$, there is $\rho_*(a) > 0$ such that for each fixed $0 < \rho < \rho_*(a)$ there is a fixed temperature $T = T(a, \rho) > 0$ for which*

$$\liminf_{\substack{L \rightarrow \infty \\ N/L^3 \rightarrow \rho}} \frac{\langle u_{0,L}, \gamma_{N,L,T}^{(1)} u_{0,L} \rangle}{N} > 0.$$

The exclusion distance, density, and temperature are held fixed in this limit.

The conclusion concerns the exact canonical Gibbs state. The temperature obtained by the proof is small and is not intended to locate the phase transition. Its role is to provide a strictly positive temperature at every sufficiently small fixed gas parameter ρa^3 .

1.2 Historical context and related methods

The statistical origin of Bose–Einstein condensation is Bose’s counting of light quanta and Einstein’s extension to material particles [9, 16, 17]. For interacting particles, Penrose and Onsager formulated condensation through a macroscopic eigenvalue of the one-particle density matrix [37]. In a translation-invariant gas, the constant orbital is the natural candidate. Bogoliubov’s theory expands about its macroscopic occupation to obtain the low-energy excitation spectrum [8]; the hard-sphere calculation of Lee, Huang and Yang predicts the leading energy correction and depletion [27]. Establishing the occupation for the exact equilibrium state is a separate mathematical question from deriving these expansions.

For hard spheres, rigorous energy bounds began with Dyson [15], and Lieb and Yngvason obtained the leading dilute asymptotics [31]. Fournais and Solovej proved the Lee–Huang–Yang lower bound for interactions including hard cores [21]; Basti, Brooks, Cenatiempo, Olgiati and Schlein proved the matching upper bound for hard spheres and then for general nonnegative short-range interactions [4, 5]. At positive temperature, Fournais, Junge, Girardot, Morin, Olivieri and Triay obtained a low-temperature free-energy lower bound that includes hard cores [19]. These results control thermodynamic energies. They do not by themselves give a volume-uniform lower bound on a single orbital’s occupation: the one-particle kinetic gap tends to zero as the box grows.

Rigorous continuum condensation has been established in several scaling regimes. Lieb and Seiringer proved complete ground-state condensation in the trapped Gross–Pitaevskii limit [30]. At positive temperature, Deuchert, Seiringer and Yngvason treated a trapped gas in a combined thermodynamic and Gross–Pitaevskii limit [14], and Deuchert and Seiringer treated the homogeneous canonical gas in Gross–Pitaevskii scaling [12]. The latter theorem allows hard cores, but their scattering length is proportional to L/N and therefore vanishes at fixed density as N grows. Further results of Fournais, Chong, Liang and Nam, and Junge allow increasing box sizes coupled to increasing dilution [10, 20, 26]. Such coupled limits do not permit unrestricted volume growth at a fixed positive gas parameter. Deuchert and Seiringer also established positive-temperature condensation in a combined semiclassical and mean-field limit for a trapped gas [13].

Fixed-parameter thermodynamic condensation also has precedents with different interactions and state formulations. Aizenman, Lieb, Seiringer, Solovej and Yngvason proved low-temperature

condensation for a hard-core lattice gas at half filling [2]. Sütő’s continuum theorem assumes both the potential and its Fourier transform to be nonnegative, with integrability and regularity hypotheses that exclude a literal hard core [39, Theorem 1.2]. Galanda and Pinamonti construct algebraic equilibrium states for smooth interactions in coupled weak-interaction limits and, after renormalization, in Gross–Pitaevskii scaling with vanishing temperature [22, 23]. The present theorem concerns the exact canonical hard-sphere state with the exclusion distance, positive density, and positive temperature all fixed before the volume grows. It gives a positive condensate fraction at sufficiently low temperature; it does not determine the transition temperature or the leading depletion.

Two companion articles provide the immediate starting point. *Ground-state condensation in the dilute hard-sphere gas* [36] treats the fixed-parameter ground-state problem and develops hard-core bridge replacements, movable groups of particles, routes avoiding unusable cells, and cancellation of conditional normalizers. *A density-uniform condensate bound for dilute Bose gases* [35] treats bounded finite-range repulsive potentials at positive temperature and supplies localization and short-slab trajectory estimates, as well as the underlying lattice-path construction. We reproduce the arguments needed here. The additional issue is to obtain joint spatial and trajectory estimates for the hard-core canonical Gibbs state, and to make the short-path comparison control its one-particle density matrix.

The local analysis builds on established tools. The hard-core ray estimate belongs to the Dyson local-energy method, and the subdivision and two-particle bounds are related to the work of Lieb and Yngvason and of Lundholm, Portmann and Solovej [15, 31, 34]. Particle insertion has antecedents in the nearest-neighbor trial states of Dyson and Lieb, Seiringer and Yngvason [15, 32]. Localization into two particle species uses the Fock-space localization formalism of Dereziński and Gérard, its many-body development by Lewin, and its thermal application by Deuchert, Seiringer and Yngvason [11, 14, 28]. Strong subadditivity and monotonicity of relative entropy provide the entropy comparisons [29, 33]. The use of positive imaginary-time paths goes back to Feynman’s formulation [18]; Ueltschi’s account distinguishes cycle information from one-particle off-diagonal order [41]. Here the overlap identity addresses the one-particle occupation directly.

The long-distance comparison has a separate probabilistic ancestry. Benjamini, Pemantle and Peres constructed paths with exponential intersection tails [6]; Häggström and Mossel developed a construction using random biases on blocks at several scales [25]. The finite-endpoint averaging method of Abbe, Massoulié, Montanari, Sly and Srivastava, and its adaptation by Garban and Spencer, turn intersection control into long-range information or order [1, 24]. Averaging path likelihoods before taking their second moments also appears in trail detection and in the point-reassignment argument of Peres and Sly [3, 7, 38]. Our lattice construction follows the explicit skeleton in OpenAI [35, Section 8.1] and the obstacle detours in OpenAI [36, Section 7], using Timár’s boundary-connectivity theorem [40, Theorem 4]. The hard-core comparison retains the exact interaction normalizers; after averaging two route changes, their cost is controlled by mutual encounters rather than by the length of either route.

1.3 The argument

The starting point is the short-path comparison developed for the hard-sphere ground state in OpenAI [36]. Its use at positive temperature requires two changes. First, the local hypotheses must hold jointly for arbitrarily many prescribed cells under a thermal path measure. Second, the path comparison must control the one-particle density matrix of a mixed state. We prove the needed thermal estimates and retain the deterministic comparison mechanism. The bridge, lattice, and measure comparison arguments used below are included with proofs.

To identify the overlap we must bound, let $\Phi(X, W')$ be the nonnegative integral kernel of $\Gamma_{N,L,T}^{1/2}$, with $X, W' \in \Lambda_L^N$. Write $X = (x, X')$, where X' is the ordered configuration of the other $N - 1$ particles, called the bath. Factoring the Gibbs kernel through its square root gives

$$\frac{\langle u_{0,L}, \gamma_{N,L,T}^{(1)} u_{0,L} \rangle}{N} = \frac{1}{L^3} \int \Phi(x, X', W') \Phi(y, X', W') \, dx \, dy \, dX' \, dW'.$$

Thus the two factors have the same bath X' and retained configuration W' , while the distinguished particle moves from x to y . Section 5 realizes $\Phi(X, W')^2$ as the joint law of the midpoint configuration and W' on a short Brownian time interval. Once the interval endpoints and W' are fixed, the conditional law is a product of Brownian bridge laws conditioned on hard avoidance. This representation lets us analyze local changes against deterministic obstacle paths while retaining the mixed-state overlap above.

The estimates are made in the scaled units of Section 2, in which a unit cell contains $D \gg 1$ particles and the core radius is α/D , with α in a fixed interval. Sections 3 and 4 give joint bounds for low cell counts and excessively close neighbors. These estimates use entropy as well as energy: a relative-entropy inequality controls a diagonal change of density, and a family of normalized insertion maps with a uniform frame bound controls the free-energy cost of adding a particle, even in bounded attractive fields.

The remaining local conditions concern whole trajectories. Section 5 marks paths that violate a condition, assigns auxiliary particle species to the marks, and removes their hard-core constraints in an upper bound. A trace estimate for a heat semigroup between two Lipschitz multipliers converts the marked path moments into free-energy estimates; exterior powers permit control of every positive trace power. These arguments adapt the thermal methods of OpenAI [35] to Dirichlet hard-core domains. They yield an exponential bound in the number of prescribed defective cells. This joint bound will control obstructing clusters when routes are drawn through the remaining cells.

The local bridge estimates of Section 6 show that most cells contain many labels whose paths can be moved through any adjacent cell with uniformly positive success probability. Section 7 splits these labels into groups and exposes all paths outside one group. The usable cells are then determined by this exposed environment. Section 8 constructs random routes between many pairs of cells, with a bounded exponential moment for the encounters of two independently drawn routes in the same environment.

Section 9 uses one such route to construct two path configurations with shifted midpoint assignments. Removing a different tagged particle from each leaves the same bath midpoints and the same W' , as required by the overlap identity. To compare the resulting measure with the two original midpoint laws, we average over routes before taking the second moment of its density. Conditional normalizers cancel away from encounters between two routes, so the cost is controlled by their encounters rather than by their lengths. The lattice estimate bounds that cost, and a common-measure inequality gives positive overlap for a positive fraction of all cell pairs.

The order of the choices is essential. The path duration and the group size are fixed before the density threshold, and all are fixed before the volume tends to infinity. Section 10 returns to physical units and verifies directly that the resulting temperature is independent of that volume.

2 A scaled formulation

The proof separates the thermodynamic limit from the small gas parameter. We first work in units in which each unit cell contains many particles, while the exclusion distance is much smaller than their typical separation. The physical change of units will be made in Section 10.

Let $K \geq 100$ be an integer and let $\Lambda_K = (\mathbb{R}/K\mathbb{Z})^3$. Write

$$V = K^3, \quad D = N/V, \quad r = \alpha/D, \quad \alpha \in [\alpha_0/2, 2\alpha_0], \quad 0 < T \leq D^{-1}, \quad \beta = T^{-1}. \quad (2.1)$$

The number $\alpha_0 \geq 2$ will be fixed sufficiently large; subsequently D will be taken sufficiently large. For $j \geq 1$, let

$$\Omega_j = \{X \in \Lambda_K^j : d(x_i, x_k) > r \text{ for } i \neq k\}, \quad q_j[f] = \sum_{i=1}^j \int_{\Omega_j} |\nabla_i f|^2.$$

Here d is Euclidean torus distance. The form domain is $H_0^1(\Omega_j)$, restricted to symmetric functions for the bosonic Hamiltonian H_j . Functions are extended by zero to Λ_K^j . This convention is used for partial traces, multipliers, and integral kernels throughout the paper.

For a bounded measurable function $W \geq 0$, set

$$H_j^{(-W)} = H_j - \sum_{i=1}^j W(x_i), \quad Z_j^{(-W)} = \text{Tr } e^{-\beta H_j^{(-W)}}, \quad F_j^{(-W)} = -T \log Z_j^{(-W)}.$$

The trace is on the symmetric space. Omit the superscript when $W = 0$, and put $Z_0 = 1$, $F_0 = 0$. For a symmetric trace-one state σ of finite kinetic energy, write

$$S(\sigma) = -\text{Tr } \sigma \log \sigma, \quad \mathcal{F}_j^{(-W)}(\sigma) = \text{Tr } H_j^{(-W)} \sigma - TS(\sigma).$$

Energy expectations mean form expectations. The entropy bound in Lemma 3.3 implies that these entropies are finite. Gibbs' variational principle gives $\mathcal{F}_j^{(-W)}(\sigma) \geq F_j^{(-W)}$. The insertion construction in Section 3 also verifies that the relevant hard-core form domains, through particle number $2N$, are nonzero for the parameters used here.

The unit cells are $B_v = v + [-1/2, 1/2]^3$, periodically interpreted, with $v \in \mathcal{L}_K = (\mathbb{Z}/K\mathbb{Z})^3$. The function $n_v(X)$ counts positions in B_v . For a set of cells $A \subset \mathcal{L}_K$, distances to A mean distances to its centers. Changing between the union of cells and their centers changes the estimates below only by absolute constants. The lattice distance d_∞ is the periodic maximum distance on $\mathcal{L}_K$. Unless stated otherwise, $c, C > 0$ denote constants independent of K, D, T in the ranges under consideration; they may change from line to line and depend on parameters already fixed. Dependence on the fixed interval for α is allowed after α_0 is chosen.

2.1 Positive kernels and retained particles

Let H_j^{lab} be the labelled Dirichlet operator on Ω_j , let $K_s^{(j)}$ be its heat kernel, and let P_j be the orthogonal projection onto symmetric functions. When $j = N$, denote the kernel of $e^{-sH_N^{\text{lab}}} P_N$ by B_s^{sym} . This operator is the bosonic semigroup extended by zero on the orthogonal complement of the symmetric subspace. Explicitly it is the average of $K_s^{(N)}$ over permutations of one endpoint. It is nonnegative and obeys the semigroup law with integration over ordered configurations, without any additional factorial.

The labelled kernel is the free Brownian kernel, with generator Δ , restricted to paths whose pair distances stay strictly above r . One way to establish this exact boundary convention is to exhaust Ω_j by relatively compact smooth open sets. Their Dirichlet forms increase in domain to $H_0^1(\Omega_j)$, giving strong semigroup convergence. A continuous path that stays in Ω_j on a compact time interval has compact image there and is captured by the exhaustion. Monotone convergence of killed path

measures proves the representation. The same reasoning, or bounded-potential Feynman–Kac, treats the fields above. This is the open-domain argument used in OpenAI [36, Section 4].

Heat traces are finite: the zero-extension form embedding and the min–max principle compare the Dirichlet eigenvalues to those of the free torus operator. Kernel identities are needed only almost everywhere. Splitting a semigroup at positive times justifies diagonal trace formulas without selecting values of a general kernel on a null set.

Partial traces preserve the remaining hard-core conditions and the kinetic expectations of the retained coordinates. The form-domain justification is given with the entropy estimates in Section 3, before these marginals are used to compare particle numbers.

2.2 The estimate to be proved

Let $\Gamma_N = Z_N^{-1} e^{-\beta H_N}$, let $\gamma_N^{(1)} = N \operatorname{Tr}_{2,\dots,N} \Gamma_N$, and let $u_0 = V^{-1/2}$. The occupation fraction is

$$B_{\text{occ}} = N^{-1} \langle u_0, \gamma_N^{(1)} u_0 \rangle.$$

Theorem 2.1 (Scaled occupation bound). *There are constants $\alpha_0 \geq 2$, $D_* < \infty$, and $c_* > 0$ such that $B_{\text{occ}} \geq c_*$ whenever the parameters satisfy (2.1), $D \geq D_*$, and K is sufficiently large. The lower threshold for K can be chosen locally uniformly for D in $[D_*, \infty)$, uniformly in $\alpha \in [\alpha_0/2, 2\alpha_0]$ and $0 < T \leq D^{-1}$.*

Sections 3 and 4 establish free-energy estimates and joint bounds on the position counts. Section 5 extends those bounds to a short interval of particle paths. The remaining sections compare configurations by moving selected labels along paths of usable cells. This yields Theorem 2.1 in Section 9; the final change of units proves Theorem 1.1.

3 Local kinetic energy and particle-number costs

The spatial estimates require two complementary comparisons. Hard exclusion forces a kinetic cost when many particles occupy one cell or have a close neighbor. Conversely, inserting one particle has a uniformly bounded free-energy cost, even in an attractive external field. We establish both comparisons, including the entropy estimates that distinguish the Gibbs problem from its ground-state counterpart.

For a set I of particle labels, write

$$q_I[F] = \sum_{i \in I} \int |\nabla_i F|^2, \quad n_v^I = \#\{i \in I : x_i \in B_v\}, \quad d_i^I = \min_{j \in I \setminus \{i\}} d(x_i, x_j).$$

The last minimum is $+\infty$ for a singleton set. Superscripts I are omitted when every label is retained. All form functions below are extended by zero across the forbidden configurations.

3.1 Two local consequences of the hard core

The following arguments reproduce the local estimates of OpenAI [36, Section 3, “Two local kinetic inequalities”]. Their ingredients are the two-particle coercivity used in local exclusion estimates [34, Proposition 10] and Neumann subdivision [31].

Lemma 3.1 (Cube energy). *Let $Q \subset \mathbb{R}^3$ be a cube of side l , and let $F \in H^1(Q^n)$ vanish almost everywhere whenever some pair has distance at most r . No boundary condition is imposed on the outer faces. Then*

$$\sum_{i=1}^n \int_{Q^n} |\nabla_i F|^2 \geq c_{\text{cube}} \frac{r}{l^3} n(n-1) \int_{Q^n} |F|^2, \quad c_{\text{cube}} = \frac{1}{64\sqrt{3}}. \quad (3.1)$$

Consequently, for a torus hard-core form function and every fixed label set I ,

$$q_I[F] \geq c_{\text{cube}} r \int \sum_v n_v^I (n_v^I - 1) |F|^2. \quad (3.2)$$

Proof. Fix a point $y \in Q$. A slice f vanishing on $|x - y| \leq r$ satisfies on almost every ray $x = y + s\omega$ in Q

$$|f(y + s\omega)|^2 \leq \left(\int_r^s t^{-2} dt \right) \int_r^s |\partial_t f(y + t\omega)|^2 t^2 dt \leq r^{-1} \int_r^{R(\omega)} |\partial_t f|^2 t^2 dt,$$

where convexity gives $R(\omega) \leq \sqrt{3}l$. Rays ending before r contribute zero. Integration in s and ω yields

$$\int_Q |f|^2 \leq \frac{\sqrt{3}l^3}{r} \int_Q |\nabla f|^2. \quad (3.3)$$

Sobolev slicing, or approximation on the ray intervals, justifies this calculation for H^1 functions.

For $n \geq 2$, subdivide Q into j^3 cubes, with $j = \lfloor (n/2)^{1/3} \rfloor$. Then $n/16 \leq j^3 \leq n/2$. In a sector specifying the subcube of every coordinate, at most j^3 coordinates occupy singleton cubes. For each of the at least $n/2$ remaining coordinates, choose a partner in its own subcube by a fixed label ordering. Apply (3.3) in that coordinate, with the partner and all other coordinates fixed. Summing uses each coordinate's kinetic energy once and gives the lower bound

$$\frac{rj^3}{\sqrt{3}l^3} \frac{n}{2} \geq \frac{rn^2}{32\sqrt{3}l^3}$$

times the sector norm. Restriction to sectors introduces no derivative of their indicators: this is Neumann bracketing. Sum over sectors; the cases $n = 0, 1$ are immediate. This proves (3.1) with the stated smaller constant. Finally assign the coordinates in I to unit cells, fix the other coordinates, and apply the cube inequality inside each occupied cell. Euclidean distance at most r in a cell lift implies torus distance at most r , so the required zeros persist. Integration over the fixed coordinates proves (3.2). $\square$

Lemma 3.2 (Close-neighbor energy). *For $0 < r < s < 1$, every hard-core form function satisfies*

$$q_I[F] \geq c_{\text{near}} r s^{-3} \int \#\{i \in I : d_i^I \leq 2s\} |F|^2, \quad c_{\text{near}} = \frac{27}{262144\sqrt{3}}. \quad (3.4)$$

Proof. Partition each coordinate circle into $\lfloor K/(8s) \rfloor$ equal intervals and translate the resulting grid uniformly. Its cube side l lies in $[8s, 16s]$. Two points of torus distance at most $2s$ belong to one cube with probability at least $(1 - 2s/l)^3 \geq 27/64$. For any fixed translated grid, apply (3.3) to every coordinate having a partner in its cube, choosing the partner by label order in each assignment sector. This bounds the kinetic energy below by $r/(\sqrt{3}l^3)$ times the number of such coordinates. For each i with $d_i^I \leq 2s$, fix a nearest partner before averaging over grid translations. It belongs to the same cube with probability at least $27/64$. Since $l^3 \leq 4096s^3$, averaging proves the claim. $\square$

3.2 Entropy and admissible marginals

We shall apply kinetic inequalities to mixed states, to their marginals, and to states confined to a spatial region. The following estimate is the ideal Bose entropy bound used in OpenAI [35, Section 3, “An entropy bound”].

Lemma 3.3 (Kinetic control of entropy). *Let O be an open subset of Λ_K with Dirichlet kinetic energy, or the whole torus with periodic kinetic energy. For every fixed $a > 0$ and every symmetric trace-one j -particle state σ of finite kinetic energy,*

$$a \operatorname{Tr} \left(\sum_{i=1}^j -\Delta_{O,i} \right) \sigma - TS(\sigma) \geq -Tj - C_a T |O| (V^{-1} + T^{3/2}). \quad (3.5)$$

In particular $S(\sigma) < \infty$. The statement holds for all $T > 0$, including $j = 0$ with the vacuum convention.

Proof. The Gaussian image sum for the periodic heat kernel gives $K_t(0) \leq C(V^{-1} + t^{-3/2})$. Killing can only decrease its diagonal, and hence

$$\operatorname{Tr} e^{ta\Delta_O} \leq C_a |O| (V^{-1} + t^{-3/2}).$$

The ideal Bose grand partition function at fugacity e^{-1} has logarithm

$$\sum_{m \geq 1} \frac{e^{-m}}{m} \operatorname{Tr} e^{m\beta a \Delta_O} \leq C_a |O| (V^{-1} + T^{3/2}).$$

Its j -particle contribution is e^{-j} times the canonical partition function. The Gibbs variational inequality gives (3.5). For a general finite-energy state the same inequality follows by finite spectral approximation, or from nonnegativity of relative entropy to the ideal canonical Gibbs state. This also proves entropy finiteness without an initial entropy assumption. Hard-core states are admissible here by Dirichlet zero extension into the free-particle space. $\square$

We record a domain fact used repeatedly below. A marginal obtained by retaining some particle coordinates of a finite-energy hard-core state still has finite kinetic energy and satisfies the hard-core Dirichlet condition among the retained coordinates. Indeed, first approximate a form vector by smooth functions compactly supported in the allowed configuration set. Slicing in the discarded variables places it in the retained form domain with values in the L^2 space of the other variables. Expansion in an orthonormal basis of that space shows that the squared form norms of its coefficient functions sum to the integrated form norm. Passing to the limit and then to a spectral decomposition of the state proves the assertion for partial traces. The kinetic expectation in the retained variables is unchanged. Symmetry of a bosonic state places each such marginal on the corresponding symmetric subspace.

3.3 Insertion without changing the old marginal

The nearest-neighbor trial construction goes back to Lieb et al. [32, Section 3.2]. We use the configurationwise normalization of OpenAI [36, Section 3, “Normalized insertion”], because it preserves the old marginal exactly.

Lemma 3.4 (Normalized insertion). *Let w be a real normalized smooth one-particle orbital and F an n -particle hard-core form function. Suppose that on the support of F at most k old positions lie within distance $2r$ of $\text{supp } w$. If $C\|w\|_\infty^2 kr^3 \leq 1/4$, there is an admissible $(n+1)$ -particle function $\tilde{F}$ of the same norm, with the same old-coordinate position marginal, such that*

$$q_{n+1}[\tilde{F}] \leq q_n[F] + C(\|\nabla w\|_2^2 + \|w\|_\infty^2 kr)\|F\|_2^2. \quad (3.6)$$

No symmetry in the new coordinate is asserted. Symmetry among the old coordinates is preserved whenever F has it.

Proof. Fix a nondecreasing Lipschitz cutoff h_r with values in $[0, 1]$, zero up to $1.1r$, one beyond $2r$, and Lipschitz constant at most C/r . Set

$$f_X(y) = w(y) \min_{1 \leq i \leq n} h_r(d(y, x_i)), \quad z(X) = \int f_X(y)^2 dy.$$

An empty minimum equals one. The relevant radius- $2r$ balls have total volume at most Ckr^3 , so $z \geq 3/4$ on the support under consideration. Almost everywhere, differentiation of the minimum requires only a minimizing old coordinate and y . Thus

$$\int \left(|\nabla_y f_X|^2 + \sum_i |\nabla_i f_X|^2 \right) dy \leq C(\|\nabla w\|_2^2 + \|w\|_\infty^2 kr). \quad (3.7)$$

Here a cutoff derivative has squared size at most C/r^2 and is supported in those balls; ties cause no change to the almost-everywhere Lipschitz derivative bound.

Where $z > 1/4$, put $a_X = f_X/\sqrt{z}$. For an old-coordinate derivative,

$$\partial a_X = z^{-1/2}(\partial f_X - a_X \langle a_X, \partial f_X \rangle_{L^2(dy)}).$$

Orthogonal projection therefore bounds its squared $L^2(dy)$ norm by $z^{-1}\|\partial f_X\|_2^2$. The new-coordinate derivative has the same bound, since z is independent of y . Furthermore

$$\int a_X^2 dy = 1, \quad \int a_X \nabla_i a_X dy = 0.$$

For $\tilde{F}(X, y) = F(X)a_X(y)$ the second identity cancels every old kinetic cross term after integrating in y . The first preserves the norm and old marginal, and (3.7) proves (3.6).

To define the multiplier globally replace $\sqrt{z}$ by $\max(\sqrt{z}, 1/2)$. This changes none of the integrals on the support of F ; a Sobolev function and its weak gradient vanish almost everywhere on its zero set. The extra clearance $1.1r$, together with compactly supported approximation of F , places the result in the enlarged Dirichlet form domain. This proves the assertion for arbitrary form functions, not only smooth ones. $\square$

Proposition 3.5 (Particle-number free-energy bounds). *For sufficiently large D , uniformly in $K \geq 100$, $0 < T \leq D^{-1}$, and the fixed α window,*

$$F_N \geq c\alpha N, \quad (3.8)$$

$$F_{N+k} \geq F_N + c\alpha k \quad (0 \leq k \leq N), \quad (3.9)$$

$$F_j^{(-W)} - F_{j-1}^{(-W)} \leq C\alpha \quad (1 \leq j \leq 2N). \quad (3.10)$$

The last estimate holds for every bounded measurable $W \geq 0$, with a constant independent of W . All form domains through particle number $2N$ are nonempty. The constants c, C can be numerical; only the large- D threshold depends on the fixed window for α .

Proof. First, all the form domains through particle number $2N$ are nonempty. Use Lemma 3.4 with $w = V^{-1/2}$, starting from the vacuum. At every step its smallness condition is bounded by $C(2N)r^3/V \leq C\alpha^3/D^2$, which is small for large D . Taking the modulus of the resulting nonzero labelled vector and symmetrizing gives a nonzero bosonic form vector. Bounded fields do not alter the form domain.

Reserve half the kinetic energy for Lemma 3.3 and apply Lemma 3.1 to the other half. Since $\sum_v n_v(n_v - 1) \geq N(D - 1)$, every N -particle state satisfies

$$\mathcal{F}_N(\sigma) \geq \frac{1}{2}c_{\text{cube}}rN(D - 1) - TN - CT(1 + VT^{3/2}).$$

With $T \leq D^{-1}$ and $\alpha \geq 1$, the negative terms are absorbed by a fixed fraction of αN for large D . This proves (3.8).

As in OpenAI [35, Section 3, “Particle-number free-energy bounds”], F_j/j is nondecreasing wherever the form domains are nonempty. For a symmetric j -particle state let S_i be the entropy of its trace-one i -particle marginal, with $S_0 = 0$. Strong subadditivity [29] gives $2S_i \geq S_{i-1} + S_{i+1}$; at $i = 1$ ordinary subadditivity suffices. All these entropies are finite by Lemma 3.3 and the preceding marginal-domain argument. Thus S_i/i is nonincreasing in i . The kinetic energy per particle is unchanged under passage to a marginal. Applying the variational principle to that admissible symmetric marginal proves monotonicity of F_j/j . In particular $F_{N+k} \geq (N+k)F_N/N$, which yields (3.9).

For the upper bound, first distinguish the last label and impose symmetry only among the first $j - 1$ coordinates. Take $w = V^{-1/2}$ in Lemma 3.4. Uniformly for $j \leq 2N$, the loss of norm before normalization is at most $Cjr^3/V \leq C\alpha^3/D^2$. For large D , the normalized orbital therefore satisfies

$$|a_X(y)| \leq CV^{-1/2}, \quad \int \left(|\nabla_y a_X|^2 + \sum_i |\nabla_i a_X|^2 \right) dy \leq C\alpha. \quad (3.11)$$

For $p \in (2\pi/K)\mathbb{Z}^3$ define the isometry

$$v_p F(X, y) = F(X) a_X(y) e^{ip \cdot y}.$$

The old kinetic cross terms vanish by normalization, and the new phase adds exactly $|p|^2 \|F\|_2^2$, since a_X is real. Old field expectations are unchanged, while the new field is nonpositive. Hence the form-energy increase, including the field, is at most $(C\alpha + |p|^2) \|F\|_2^2$.

These insertions form a bounded frame:

$$\sum_{|p| \leq 1} v_p v_p^* \leq CI. \quad (3.12)$$

Indeed, after zero extension in y , Parseval’s identity at each X gives $\sum_p |\int a_X(y) e^{-ip \cdot y} u(X, y) dy|^2 = V \int a_X(y)^2 |u(X, y)|^2 dy \leq C \int |u(X, y)|^2 dy$. Integrate in X and restrict the frequency sum to obtain (3.12).

Let $Z_{j, \text{dist}}^{(-W)}$ denote the heat trace in this distinguished-label space, and let E_l, ψ_l be an orthonormal eigenbasis of the symmetric $(j - 1)$ -particle operator with field. The scalar Jensen inequality in the spectral measure of each inserted unit vector gives

$$\begin{aligned} CZ_{j, \text{dist}}^{(-W)} &\geq \sum_{|p| \leq 1} \sum_l \langle v_p \psi_l, e^{-\beta H_{j, \text{dist}}^{(-W)}} v_p \psi_l \rangle \\ &\geq \sum_{|p| \leq 1} \sum_l e^{-\beta(E_l + C\alpha + |p|^2)} \geq cV e^{-\beta(C\alpha + 1)} Z_{j-1}^{(-W)}. \end{aligned}$$

The first inequality is the frame bound; the last uses at least cV frequencies with $|p| \leq 1$. Positivity of the labelled heat kernel gives

$$Z_j^{(-W)} \geq j^{-1} Z_{j,\text{dist}}^{(-W)} :$$

in the permutation expansion of the full symmetrizer retain the $(j-1)!$ permutations fixing the last label, whose contributions are nonnegative. Therefore

$$F_j^{(-W)} - F_{j-1}^{(-W)} \leq C\alpha + 1 + T \log_+(Cj/V) \leq C'\alpha,$$

because $j/V \leq 2D$, $T \leq D^{-1}$, and $\alpha \geq 1$. □

4 Localization and joint spatial estimates

We now show that any prescribed family of cells is unlikely to suffer simultaneous particle shortages or simultaneous failures of separation. These are estimates under the actual equilibrium position law μ . They do not condition on the positions outside the cells. The method, adapted from OpenAI [35, Section 4], compares a lower free-energy penalty from spatial localization with an upper bound obtained by tilting the diagonal of the Gibbs state.

For a bounded measurable one-particle function f , put $n_f(X) = \sum_i f(x_i)$, and write $\mathbb{E}_\sigma$ for expectation under the position law of a state σ . In this section set

$$\Gamma = Z_N^{-1} e^{-\beta H_N}, \quad \mu(dX) = \Gamma(X, X) dX.$$

4.1 Localization and diagonal tilting

Lemma 4.1 (Two-color localization). *Let p, q be real Lipschitz functions with $p^2 + q^2 = 1$, and suppose p is compactly supported in an open set O , or take $O = \Lambda_K$. For a finite-energy symmetric J -particle state σ , map each coordinate by $f \mapsto (pf, qf)$ and dephase by the number j of particles in the first color. Denote its distribution by w_j and the conditional first- and second-color marginal states by $\sigma_{i,j}$ and $\sigma_{o,j}$. For every bounded field $W \geq 0$,*

$$\mathcal{F}_J^{(-W)}(\sigma) + \mathbb{E}_\sigma n_{|\nabla p|^2 + |\nabla q|^2} \geq \sum_{j=0}^J w_j [\mathcal{F}_j^{(-W)}(\sigma_{i,j}) + \mathcal{F}_{J-j}^{(-W)}(\sigma_{o,j})] - TH(w). \quad (4.1)$$

The inside states also satisfy the Dirichlet condition in O . Terms with $w_j = 0$ are omitted, and vacuum states are allowed. Moreover

$$\sum_j jw_j = \mathbb{E}_\sigma n_{p^2}, \quad H(w) \leq -\log(1 - e^{-1}) + \sum_j jw_j. \quad (4.2)$$

Proof. The geometric localization and entropy decomposition are the many-body IMS construction; see Lewin [28, Section 3] and Deuchert et al. [14, Section 3, Lemma 3.1]. Here is the hard-core version. The one-particle map is isometric, so its symmetric tensor power preserves the spectrum and entropy of σ . In each sector of fixed color number, normalized symmetrization over the mutually orthogonal color assignments identifies the space with the tensor product of the two symmetric species spaces. There is one such tensor product, with no binomial multiplicity.

The product rule and $p\nabla p + q\nabla q = 0$ give the exact kinetic increase $\mathbb{E}_\sigma n_{|\nabla p|^2 + |\nabla q|^2}$. The field distributes between the two colors without error. The masks preserve the hard-core form domain. After cross-color exclusions are dropped, slicing gives admissible same-color marginal states, by the domain argument in Section 3; the first color is also Dirichlet in O . Thus the sum of their

energy expectations equals the localized energy. Entropy cannot decrease under dephasing and is subadditive in each color sector, whence

$$S(\sigma) \leq H(w) + \sum_j w_j [S(\sigma_{i,j}) + S(\sigma_{o,j})].$$

All entropies are finite by Lemma 3.3, applied also to the finitely many color spaces. This proves (4.1). Conditional on positions, colors are independent choices with probabilities $p(x_i)^2, q(x_i)^2$, proving the first identity in (4.2). Nonnegativity of relative entropy to the geometric law $(1 - e^{-1})e^{-j}$ proves the second. $\square$

Lemma 4.2 (Diagonal tilt). *For a bounded real symmetric Lipschitz function G of N positions, set*

$$g = e^G, \quad M_G = \mathbb{E}_\mu e^{2G}, \quad \sigma_g = M_G^{-1} g \Gamma g.$$

Then

$$\mathcal{F}_N(\sigma_g) - F_N \leq \mathbb{E}_{\sigma_g} |\nabla G|^2 + T(2\mathbb{E}_{\sigma_g} G - \log M_G). \quad (4.3)$$

Here $|\nabla G|^2 = \sum_i |\nabla_i G|^2$.

Proof. This is the diagonal-tilting argument of OpenAI [35, Section 4, “Two-color localization and diagonal tilting”]. Put $\tau = M_G^{-1} \Gamma^{1/2} g^2 \Gamma^{1/2}$. The nonzero spectra of τ and σ_g agree, as the spectra of $B^* B$ and $B B^*$ for $B = M_G^{-1/2} g \Gamma^{1/2}$. If $H_N \psi = E \psi$, testing the weak equation with $g^2 \psi$ gives

$$q_N[g\psi] = E \|g\psi\|_2^2 + \int |\nabla g|^2 |\psi|^2.$$

Lipschitz multiplication preserves the hard-core boundary condition. Summing in a Gibbs eigenbasis yields

$$\text{Tr } H_N \sigma_g = \text{Tr } H_N \tau + \mathbb{E}_{\sigma_g} |\nabla G|^2. \quad (4.4)$$

Both energies are finite: g and its gradients are bounded at fixed N, K , and positive-time heat traces control the eigenvalue-weighted sum. Lemma 3.3 then gives finite entropies.

To bound the relative entropy $D(\tau \| \Gamma)$, first let g^2 be constant on a finite partition into symmetric measurable sets E . Put $p_E = \text{Tr } \Gamma \mathbf{1}_E$ and $\tau_E = p_E^{-1} \Gamma^{1/2} \mathbf{1}_E \Gamma^{1/2}$ when $p_E > 0$. The two states Γ and τ are mixtures of the same states τ_E , with weights p_E and $p_E g_E^2 / M_G$, respectively. Monotonicity of relative entropy under discarding the classical label [33] gives

$$D(\tau \| \Gamma) \leq \sum_E \frac{p_E g_E^2}{M_G} \log \frac{g_E^2}{M_G}.$$

Uniform positive simple approximation to g^2 gives trace-norm convergence of these states and convergence of the classical expression. Lower semicontinuity therefore implies $D(\tau \| \Gamma) \leq 2\mathbb{E}_{\sigma_g} G - \log M_G$. Finally $\mathcal{F}_N(\tau) - F_N = T D(\tau \| \Gamma)$, the equality of entropies, and (4.4) prove (4.3). $\square$

4.2 A local penalty and density control in a field

For a nonempty set of sites $A \subset \mathcal{L}_K$, define, for a configuration of any particle number $J \leq N$,

$$s_0 = b D^{-1/3}, \quad 0 < b < 1, \quad (4.5)$$

$$f_A(X) = \sum_{i=1}^J e^{-d(x_i, A)}, \quad (4.6)$$

$$k_A(X) = \#\{i : d(x_i, A) \leq 6, d_i \leq 2s_0\}. \quad (4.7)$$

The parameter b will be chosen small after the low-count threshold has been fixed. We always increase D so that $r < s_0 < 1/2$.

Proposition 4.3 (Weighted local free-energy penalty). *There are $c, C > 0$, independent of b , such that for every $0 \leq J \leq N$, every bounded measurable $0 \leq W \leq 1$, and every finite-energy symmetric trace-one state σ ,*

$$\mathcal{F}_J^{(-W)}(\sigma) - F_J^{(-W)} \geq \alpha \left[\mathbb{E}_\sigma(f_A + cb^{-3}k_A) - CD|A| \right]. \quad (4.8)$$

The threshold on D may depend on b , but the displayed constants do not.

Proof. We combine the exponential localization of OpenAI [36, Section 3, “Exponential spatial localization”] with the thermal entropy decomposition of Lemma 4.1. For $h \geq 0$, choose sine-cosine masks p_h, q_h such that $p_h = 1$ at distance at most h from A , $p_h = 0$ from distance $h+1$ onward, and

$$|\nabla p_h|^2 + |\nabla q_h|^2 \leq C \mathbf{1}_{\{h < d(x, A) < h+1\}}.$$

Take the inside region O_h to be the open $(h+2)$ -neighborhood, or the torus when appropriate. Its volume is at most $C(h+4)^3|A|$, and p_h is compactly supported there.

Apply Lemma 4.1. Outside, the upper increments of Proposition 3.5 give $F_{J-j}^{(-W)} \geq F_J^{(-W)} - C\alpha j$. Inside, divide the kinetic energy into three fixed positive portions. One portion pays for the entire von Neumann entropy by Lemma 3.3. Apply Lemmas 3.1 and 3.2 to the other two, with $s = s_0$. The field costs at most j , while the color entropy costs at most $T(C+j)$ by (4.2). Thus all losses linear in the inside number are bounded by $C\alpha j$, since $T \leq 1$ and $\alpha \geq 1$. The remaining entropy error is at most

$$CT[1 + |O_h|(V^{-1} + T^{3/2})] \leq C(h+4)^3|A|.$$

Average with the probability density $e^{-h} dh$. For every position x ,

$$\int_0^\infty p_h(x)^2 e^{-h} dh \leq e e^{-d(x, A)}, \quad \int_0^\infty \mathbf{1}_{\{h < d(x, A) < h+1\}} e^{-h} dh \leq e e^{-d(x, A)}.$$

Consequently the averaged removal and localization costs are at most $C\alpha \mathbb{E}_\sigma f_A$, and the residual entropy cost is at most $C\alpha|A|$.

To retain the positive terms, set $b_v = e^{-d(v, A)}$. If $h \geq d(v, A) + 1$, every coordinate in B_v belongs to the first color with probability one. The averaged cube contribution is therefore at least $cr \mathbb{E}_\sigma \sum_v b_v n_v (n_v - 1)$. A particle counted by k_A has a partner within $2s_0 < 1$; both lie at distance less than 7 from A . For $h \geq 8$ both labels belong to the first color, so its close-neighbor inequality counts this particle. This yields at least $crs_0^{-3} \mathbb{E}_\sigma k_A = cab^{-3} \mathbb{E}_\sigma k_A$. The color interpretation used here follows from the conditional independent choices in Lemma 4.1.

We have proved

$$\mathcal{F}_J^{(-W)}(\sigma) - F_J^{(-W)} \geq \mathbb{E}_\sigma \left[cr \sum_v b_v n_v (n_v - 1) - C\alpha f_A + cab^{-3} k_A \right] - C\alpha|A|.$$

Within a unit cell, distances to A differ from the center distance by less than one. Hence

$$f_A \leq e \sum_v b_v n_v, \quad \sum_v b_v \leq \sum_{w \in A} \sum_v e^{-d(v, w)} \leq C|A|.$$

The final lattice sum is uniformly bounded on periodic unit grids. For $n \geq 0$ and $D \geq 1$, completing a scalar square gives

$$\frac{c}{D} n(n-1) - (C+1)en \geq -C'D.$$

Multiply by αb_v and sum. Since $r = \alpha/D$, this absorbs all linear losses while leaving αf_A and proves (4.8). The vacuum case is immediate. $\square$

Corollary 4.4 (Density bound in a small field). *In the J -particle equilibrium state with $0 \leq J \leq N$ and $0 \leq W \leq 1$, every set A of sites satisfies*

$$\mathbb{E} n_{\mathbf{1}_{\bigcup_{v \in A} B_v}} \leq CD|A|. \quad (4.9)$$

Proof. The left side of (4.8) is zero in equilibrium. Drop the nonnegative close-count term and use $n_{\mathbf{1}_{\bigcup_{v \in A} B_v}} \leq ef_A$. Empty A and the vacuum cause no exception. $\square$

The mean estimate will control the effect of small fields on partition functions. To produce many movable particles, we need the stronger joint statements proved next.

4.3 Simultaneous shortages are exponentially unlikely

Proposition 4.5 (Joint low-count bound). *One can choose α_0 sufficiently large and then $\ell > 0$ sufficiently small so that, for every set of sites A and all sufficiently large D ,*

$$\mu\{n_v < \ell D \text{ for every } v \in A\} \leq e^{-cD|A|}. \quad (4.10)$$

The constants and thresholds are uniform in $K \geq 100$ and $0 < T \leq D^{-1}$.

Proof. Assume $A \neq \emptyset$. In each of its cells take $p = 1$ on the central cube of half-side $1/4$, with $p = 0$ outside the central cube of half-side $1/3$. Sine-cosine interpolation gives q with $p^2 + q^2 = 1$ and uniformly bounded squared gradient cost. Let O be the union of these unit-cell interiors. Choose a smooth squared cutoff $0 \leq \phi \leq 1$, supported in O , equal to one on the central cubes of half-side $1/3$, and satisfying $|\nabla \phi|^2 \leq C\phi$. In particular p^2 and the localization gradient cost are bounded by $C\phi$.

We compare each outside state from Lemma 4.1 with a state obtained by filling the empty central cubes. Set $k = \lfloor \eta D|A| \rfloor$, where $0 < \eta < 1$ will be fixed. Take a normalized smooth bump in each central cube of half-side $1/8$, and sum with equal amplitude to obtain an orbital w with

$$\|w\|_\infty^2 \leq C/|A|, \quad \|\nabla w\|_2^2 \leq C.$$

Starting at the vacuum, apply Lemma 3.4 successively k times with this orbital. Its smallness condition is satisfied because $Ckr^3/|A| \leq CDr^3 \rightarrow 0$. The resulting normalized labelled vector has energy at most

$$Ck(1 + rk/|A|). \quad (4.11)$$

It is supported in the union of the bump supports in every coordinate. Replace it by the square root of the average of its squared moduli over all permutations. The norm is unchanged, the hard-core zeros are preserved, and the inequality for the gradient of a Euclidean norm shows that energy cannot increase. This constructs a fixed symmetric normalized added vector ψ_k with the same bound. Dirichlet admissibility follows by contraction and approximation of the original form vector.

The conditional outside state is supported in $(\text{supp } q)^{N-j}$. Its coordinates are separated from every bump support by distance at least $1/8$, larger than r for large D . The map on its form vectors

$$U_j f = \binom{N-j+k}{k}^{1/2} P_{\text{sym}}(f \otimes \psi_k)$$

is an isometry: the distinct choices of the k added labels have disjoint position supports. It preserves entropy and has additive kinetic energy. The support separation also guarantees every old-new hard-core condition. Consequently

$$\mathcal{F}_{N-j}(\sigma_{o,j}) \geq F_{N-j+k} - Ck(1 + rk/|A|).$$

Here $k \leq \eta N < N$, so Proposition 3.5 applies up to $N + k \leq 2N$. Its lower increment followed by j upper increments gives

$$\mathcal{F}_{N-j}(\sigma_{o,j}) \geq F_N + c\alpha k - C\alpha j - Ck(1 + rk/|A|). \quad (4.12)$$

The bump profiles and their constants were fixed independently of α_0 . Choose α_0 large, and then η small, so that throughout the α window

$$C(1 + \alpha\eta) \leq c\alpha/2.$$

Since $rk/|A| \leq \alpha\eta$, the final term in (4.12) is absorbed by half the positive insertion gain. For large D , $k \geq \eta D|A|/2$.

Inside, Lemma 3.3 with the full kinetic energy bounds the free energy below by $-Tj - CT|A|(V^{-1} + T^{3/2})$. Apply Lemma 4.1, use $\sum_j jw_j \leq \mathbb{E}_\sigma n_\phi$, and absorb the remaining $O(|A|)$ errors into the filling gain. For every finite-energy N -particle state this gives

$$\mathcal{F}_N(\sigma) - F_N \geq c\alpha D|A| - C\alpha \mathbb{E}_\sigma n_\phi. \quad (4.13)$$

Apply Lemma 4.2 with $G = -\lambda n_\phi$, $0 \leq \lambda \leq 1$, and write $L(\lambda) = -\log \mathbb{E}_\mu e^{-2\lambda n_\phi}$. Since $0 \leq n_\phi \leq N$, this moment is differentiable at fixed N, K by dominated convergence. The squared gradient is at most $C\lambda^2 n_\phi$, while $2\mathbb{E}_{\sigma_g} G \leq 0$. Comparing with (4.13), and increasing C , gives

$$C\alpha \mathbb{E}_{\sigma_g} n_\phi + TL(\lambda) \geq c\alpha D|A|, \quad L'(\lambda) = 2\mathbb{E}_{\sigma_g} n_\phi.$$

Choose a fixed $a' > 0$ sufficiently small. Whenever $L(\lambda) \leq a'D|A|$, the term $TL(\lambda)$ is at most half the right side, and the resulting lower bound for L' is at least $2a'D|A|$. Since $L(0) = 0$ and L is nondecreasing, this forces $L(1) \geq a'D|A|$. On the low-count event, $n_\phi \leq \sum_{v \in A} n_v < \ell D|A|$. Exponential Markov therefore gives probability at most $\exp[-(a' - 2\ell)D|A|]$. Choose $2\ell < a'$ to prove (4.10). For empty A both sides equal one. $\square$

4.4 Simultaneous failures of separation

Fix from now on $\theta > 0$ so that $3\theta < \ell/2$. For a site v , define

$$C_v(X) = \#\{i : d(x_i, v) \leq 5, d_i \leq s_0\}. \quad (4.14)$$

The enlarged radius allows this count to control endpoint defects of particles starting in neighboring cells.

Proposition 4.6 (Joint close-count bound). *For every fixed $\theta > 0$, one can choose $b > 0$ sufficiently small and then D sufficiently large so that, for every set of sites A ,*

$$\mu\{C_v \geq \theta D \text{ for every } v \in A\} \leq p_D^{|A|}, \quad p_D = \exp(-c\theta b D^{2/3}) \rightarrow 0. \quad (4.15)$$

Here $c > 0$ can be numerical, and the bounds remain uniform in the temperature and volume ranges above.

Proof. For nonempty A , choose squared Lipschitz cutoffs u, ξ in $[0, 1]$, with $u = 1$ on $[0, 5]$, $u = 0$ on $[6, \infty)$, $\xi = 1$ on $[0, 1]$, and $\xi = 0$ on $[2, \infty)$. Their derivatives satisfy $|u'|^2 \leq Cu$ and $|\xi'|^2 \leq C\xi$. Define the Lipschitz enlargement

$$m_A(X) = \sum_i u(d(x_i, A))\xi(d_i/s_0).$$

For configurations with fewer than two particles take the summand with $d_i = +\infty$ to be zero. Then

$$m_A \leq k_A, \quad \sum_{v \in A} C_v \leq C_0 m_A, \quad |\nabla m_A|^2 \leq C s_0^{-2} m_A. \quad (4.16)$$

The middle inequality follows because each point lies within distance 5 of at most a fixed number C_0 of unit lattice sites, and each counted point contributes one to m_A .

For completeness, the final inequality uses the geometry of the nearest-neighbor graph, as in OpenAI [36, Section 3, “Joint count and close-neighbor tails”]. Away from a null set the nearest neighbor of each particle is unique. If two incoming edges at x_k come from x_i, x_j , with $|x_i - x_k| \geq |x_j - x_k|$, nearest-neighbor choice implies $|x_i - x_j| \geq |x_i - x_k|$. The cosine law then bounds the angle between the incoming directions below by $\pi/3$. Edges relevant to the derivatives have length at most $2s_0 < 1$, so consistent Euclidean lifts are available on our torus. Sphere packing therefore gives a numerical bound on their incoming degree. Each coordinate derivative meets only a bounded number of summands of m_A . The squared-cutoff inequalities bound the squared full gradient of each summand by $C s_0^{-2} u(d(x_i, A)) \xi(d_i/s_0)$. Bounded overlap proves (4.16).

Use Lemma 4.2 with $G = \lambda m_A$ for $0 \leq \lambda \leq s_0$. Since $M_G \geq 1$, (4.16) bounds its right side by

$$(C\lambda^2 s_0^{-2} + 2T\lambda) \mathbb{E}_{\sigma_g} m_A \leq C \mathbb{E}_{\sigma_g} m_A.$$

Proposition 4.3, with $W = 0$ and $J = N$, bounds the same free-energy excess below by $c\alpha b^{-3} \mathbb{E}_{\sigma_g} m_A - C\alpha D|A|$. Choose b small enough to absorb the upper bound. It follows that

$$\mathbb{E}_{\sigma_g} m_A \leq C b^3 D|A| \quad (0 \leq \lambda \leq s_0).$$

Since $0 \leq m_A \leq N$, differentiating under the integral and then integrating the logarithmic moment derivative gives

$$\log \mathbb{E}_\mu e^{2s_0 m_A} \leq C s_0 b^3 D|A|.$$

On the event in (4.15), $m_A \geq \theta D|A|/C_0$. Reduce b further so that $Cb^3 \leq \theta/C_0$. Exponential Markov bounds its probability by $\exp[-(\theta/C_0)s_0 D|A|]$, proving the claim because $s_0 D = bD^{2/3}$. The empty-set assertion is immediate. $\square$

The spatial choices have a fixed order: first α_0 large and ℓ small, then θ with $3\theta < \ell/2$, then b small, and finally D large. These estimates provide simultaneous endpoint control for arbitrarily many prescribed cells. The next section extends that control to every path visiting a cell during a short imaginary-time interval.

5 A thermal slab and joint trajectory estimates

The spatial estimates control the equilibrium positions. We now need the corresponding control of trajectories on a short time interval, simultaneously near any prescribed set of cells. The first step is an exact path representation of the Gibbs state that also retains the off-diagonal information needed for condensation. The second step transfers rare events for a free path to joint estimates for the interacting sample. For this transfer we adapt the auxiliary-color and trace-sandwich argument of OpenAI [35, Section “A joint bound for atypical trajectories,” in particular the lemma “Trace bound for a Lipschitz sandwich”]. The hard-core domains and the trajectory tests used here require the details given below.

5.1 The path law and its purification

Fix $0 < t \leq 1$, set $\delta = 2t$, and require $\beta > \delta$. The time t will be fixed before the lower threshold on D . Let ν_N be the measure of N independent torus Brownian motions with generator Δ on $[0, \delta]$, with Lebesgue initial measure. Write

$$X(s) = (\omega_1(s), \dots, \omega_N(s)), \quad Y = X(0), \quad Z = X(\delta).$$

In addition to these labelled paths, introduce a configuration $W' \in \Lambda_K^N$. With the symmetric heat-kernel convention of Section 2, define

$$d\mathcal{R}(\omega, W') = \frac{B_{\beta/2-t}^{\text{sym}}(W', Y) B_{\beta/2-t}^{\text{sym}}(Z, W')}{Z_N} \mathbf{1}_{\{d(\omega_i(s), \omega_j(s)) > r \ \forall s, i < j\}} d\nu_N(\omega) dW'. \quad (5.1)$$

All path spaces carry their Borel sigma fields. Continuous torus paths have continuous Euclidean lifts, unique after specifying the initial lift; lifted increments and diameters are independent of that choice.

Proposition 5.1 (Thermal purification). *Equation (5.1) defines a probability measure. At every fixed time $s \in [0, \delta]$, its position marginal is*

$$\mu(dX) = Z_N^{-1} B_{\beta}^{\text{sym}}(X, X) dX.$$

The joint law of $X(t)$ and W' has density $\Phi(X, W')^2$, where

$$\Phi(X, W') = Z_N^{-1/2} B_{\beta/2}^{\text{sym}}(X, W'). \quad (5.2)$$

This function is nonnegative, has squared integral one, and is symmetric separately in X and W' . The constant-orbital occupation fraction of the scaled Gibbs state is

$$B_{\text{occ}} = \frac{1}{V} \int \left| \int_{\Lambda_K} \Phi(x, X', W') dx \right|^2 dX' dW', \quad (5.3)$$

where $X' \in \Lambda_K^{N-1}$ is an ordered configuration.

Proof. The avoidance indicator gives the labelled Dirichlet heat kernel by the open-domain Feynman–Kac representation established in Section 2, with the same strict-avoidance and zero-extension conventions.

Let P_{sym} be the orthogonal symmetrizer in labelled configuration space. It commutes with the labelled semigroup and $P_{\text{sym}}^2 = P_{\text{sym}}$. Integrating W' in Equation (5.1) therefore leaves the outer kernel $B_{\beta-\delta}^{\text{sym}}(Z, Y)$. Composing with the slab kernel and taking the trace gives Z_N , so the total mass is one. Splitting the labelled slab kernel at time s gives the diagonal kernel $B_{\beta}^{\text{sym}}(X, X)$ at that time. These identities hold against arbitrary bounded position tests; alternatively all diagonal traces can be evaluated after a further positive-time split.

If instead we retain W' and split at t , composing the first outer kernel with the first killed half-slab gives $B_{\beta/2}^{\text{sym}}(W', X)$. The other half gives its transpose. Their product, divided by Z_N , is $\Phi(X, W')^2$.

Finally Φ is the zero-extended kernel of $\Gamma_N^{1/2}$. Thus the kernel of the trace-one Gibbs state satisfies

$$\Gamma_N(X, \bar{X}) = \int \Phi(X, W') \Phi(\bar{X}, W') dW'.$$

The reduced density matrix is N times the partial trace over X' . Its expectation in the orbital $V^{-1/2}$, divided by N , is precisely Equation (5.3). Tonelli's theorem applies since the kernels are nonnegative; the resulting integral is finite since a one-particle occupation fraction is at most one. $\square$

The extra variable W' is what permits a mixed Gibbs state to be treated by a nonnegative squared density. It will be held fixed in the later path changes. No ground-state eigenfunction enters Equation (5.1).

5.2 Regular trajectories and certificates

Recall $s_0 = bD^{-1/3}$ and the fixed parameters ℓ, θ, b from the spatial estimates, with $3\theta < \ell/2$. Put

$$h_0 = D^{-1}, \quad \gamma = \frac{1}{12}, \quad m_t = \lceil t/r^2 \rceil, \quad \tau = t/m_t. \quad (5.4)$$

Take D large enough that $h_0 < t$ and $r^2/2 \leq \tau \leq r^2$. Divide each half of the slab into m_t equal intervals, denoted together by $I_1, \dots, I_{2m_t}$. For a lifted path define

$$A_j(\omega) = r^{-1} \text{diam}(\tilde{\omega}(I_j)).$$

The following definition is the one used in OpenAI [36, Section “A stationary sample of hard-core paths,” definition of a regular path].

Definition 5.2 (Regular path). Fix a sufficiently large numerical constant C_{reg} . A path on $[0, 2t]$ is *regular* if

$$\max_j A_j \leq D^\gamma, \quad \frac{1}{2m_t} \sum_{j=1}^{2m_t} (1 + A_j)^3 \leq C_{\text{reg}}, \quad (5.5)$$

and its lifted diameter in each endpoint window $[0, h_0]$ and $[2t - h_0, 2t]$ is at most $s_0/8$.

We will use both the free-motion and the bridge forms of the next estimate. The latter supplies the individual regularity tests for the replacement paths in Section 6.

Lemma 5.3 (Free cutoff bounds). *Fix $t > 0$. For every $q > 0$, a free Brownian path on $[0, 2t]$ fails regularity with probability at most $C_{q,t}D^{-q}$, uniformly in its starting point, K , and the allowed α -window. The same assertion holds for a Euclidean bridge of duration $2t$ with bounded endpoint displacement, and for two independent Euclidean bridge legs of duration t joined at a prescribed midpoint, when their three prescribed points lie in a fixed bounded region. Constants and the lower threshold on D may depend on that region and t , but C_{reg} is a single numerical constant for all these laws. If the three points lie in the radius-3 ball about a site v , the probability that the two-leg path leaves the radius-10 ball about v is at most $Ce^{-c/t}$ for $0 < t \leq 1$.*

Proof. We reproduce the argument of OpenAI [36, Lemma “Free cutoff bounds”]. The variables A_j of Brownian motion depend on disjoint increments. Brownian scaling and $\tau/r^2 \in [1/2, 1]$ give $\mathbb{P}(A_j > a) \leq Ce^{-ca^2}$ for $a \geq 1$. Hence the maximum test fails with probability at most $Cm_te^{-cD^{2\gamma}}$.

Set $U_j = (1 + A_j)^3$. Every fixed moment of U_j is bounded uniformly. In the expansion of the $2k$ th moment of $\sum_j (U_j - \mathbb{E}U_j)$, a nonzero term uses every index at least twice, by independence and centering. At most k distinct indices can occur, so

$$\mathbb{E} \left| \frac{1}{2m_t} \sum_j (U_j - \mathbb{E}U_j) \right|^{2k} \leq C_k m_t^{-k}.$$

Choose C_{reg} larger than a fixed multiple of $1 + \sup_j \mathbb{E}U_j$. Since m_t is comparable to D^2 for fixed t and the fixed α -window, Markov’s inequality gives every prescribed inverse power of D for the

average test. The same argument applies to the average on either half separately, with the same threshold. For each endpoint window the maximal Gaussian estimate gives

$$\mathbb{P}\{\text{diam } \tilde{\omega}([0, h_0]) > s_0/8\} \leq C \exp(-cs_0^2/h_0) = C \exp(-cb^2 D^{1/3}),$$

and reversal handles the other window.

For a bridge leg, write its centered part as $B(u) - (u/t)B(t)$ and add the line joining its prescribed endpoints. On a mesh interval, the additional increment in units of r is bounded by $C(r/t)(|B(t)| + L)$, where L bounds endpoint displacement. On the event $|B(t)| \leq D^{1/12}$ this tends to zero, while the complementary probability is at most $C_t e^{-c_t D^{1/6}}$. The maximum test follows by applying the Brownian bound with threshold $D^\gamma/2$. The inequality $(x+y)^3 \leq 4x^3 + 4y^3$, with a sufficiently large fixed choice of C_{reg} , transfers the average test as well. The additional displacement on an endpoint window is at most $Ch_0(|B(t)| + L)/t = o(s_0)$ on the same event; use the Brownian window estimate with half the allowed threshold. Two independent legs require a union bound, and a duration- $2t$ bridge has the identical representation with $2t$ in place of t .

For the final assertion, each interpolating line remains in the radius-3 ball. Exiting the radius-10 ball requires a centered fluctuation of size at least seven on one leg. The Brownian representation and its maximal Gaussian bound give $Ce^{-c/t}$. $\square$

Write $C_v(X)$ for the close count from Proposition 4.6, namely

$$C_v(X) = \#\{i : d(X_i, v) \leq 5, d_i(X) \leq s_0\}.$$

A *certificate at v* consists of the following conditions on the full path sample:

1. $n_v(Y) \geq \ell D$ and $C_v(Y), C_v(Z) < \theta D$;
2. at most $H_1 D$ paths visit the radius-12 ball about v , every visiting path is regular, and fewer than θD visiting paths have lifted diameter greater than one.

These are precisely the spatial and trajectory conditions needed for the deterministic bridge estimates. In particular, the certificate is an event of the full sample; it is not information on which we will condition the paths to be changed.

Proposition 5.4 (Joint certificate supply). *For each prescribed $p_1 > 0$, one can choose H_1 large, then $t > 0$ sufficiently small, and then D_0 sufficiently large, such that, for every set A of distinct cell centers,*

$$\mathcal{R}\{\text{no site of } A \text{ has a certificate}\} \leq p_1^{|A|} \tag{5.6}$$

whenever $D \geq D_0$, $0 < T \leq D^{-1}$, and K is sufficiently large. The choices are uniform over the α -window. After H_1 is fixed, t may be decreased further before increasing D_0 .

The endpoint parts already follow from Propositions 4.5 and 4.6. The issue is to control many path failures jointly under $\mathcal{R}$. We establish that estimate next, and prove Proposition 5.4 at the end of the section.

5.3 A localized kernel for one marked path

For a nonempty set A of sites let $J_A(\omega)$ be the number of sites in A whose radius-12 balls the path visits. Consider the following three single-path events:

$$E_{\text{all}} = \{\text{all paths}\}, \quad E_{\text{diam}} = \{\text{diam } \tilde{\omega} > 1\}, \quad E_{\text{irr}} = \{\omega \text{ is not regular}\}.$$

For E equal to one of these events and $\lambda > 0$, we will bound

$$\begin{aligned}\mathfrak{M} &= \mathbb{E}_{\mathcal{R}} \exp \left(\lambda \sum_{i=1}^N J_A(\omega_i) \mathbf{1}_E(\omega_i) \right) \\ &= \mathbb{E}_{\mathcal{R}} \prod_{i=1}^N \left[1 + (e^{\lambda J_A(\omega_i)} - 1) \mathbf{1}_E(\omega_i) \right].\end{aligned}\tag{5.7}$$

Expanding this product selects subsets of marked labels, each carrying weight $(e^{\lambda J_A(\omega)} - 1) \mathbf{1}_E(\omega)$. We first bound the free-path kernel for one such marked label.

We partition paths by a nonnegative integer n . On the first half, mark the first exit from the unit ball about the starting point, then the first exit from the unit ball about that exit point, and continue up to time t . On the second half make the same construction backwards from the terminal point. Denote the total number of completed exits on the two halves by $n(\omega)$, and write n for its value on a fixed bin. Continuity and uniform continuity on the compact time interval make this number finite for each path. Between consecutive marks the lift lies in a unit ball. Thus on bin n ,

$$J_A \leq C(n+1), \quad |\tilde{\omega}(2t) - \tilde{\omega}(0)| \leq n+2.\tag{5.8}$$

If $J_A > 0$, an endpoint is at lifted distance at most $n+3$ from any visiting point: the first half's endpoint-to-midpoint displacement is at most n_1+1 , and the diameter of the other half is at most n_2+2 . Thus both endpoints have torus distance at most $n+15$ from A . By Equation (5.8), the unit tube about the straight lifted endpoint segment projects into $\{x : d(x, A) < 2n+18\}$, hence into $\{x : d(x, A) < 32(n+1)\}$.

Set $O_n = \{x : d(x, A) < 32(n+1) + 2\}$. Choose a Lipschitz cutoff $0 \leq p_n \leq 1$ equal to one when $d(x, A) \leq 32(n+1)$ and zero when $d(x, A) \geq 32(n+1) + 1$, with $|\nabla p_n| \leq C$. Its support is compact in O_n . These definitions allow O_n to be the whole torus. They satisfy

$$|O_n| \leq C(n+1)^3 |A|, \quad O_n \text{ is covered by at most } C(n+1)^3 |A| \text{ unit cells.}\tag{5.9}$$

Here $|O_n|$ is volume and $|A|$ is cardinality.

Let $\mathcal{K}_n(y, z)$ be the unnormalized free torus path kernel of duration δ with the marking weight and the bin- n restriction. Thus endpoint integration against $\mathcal{K}_n$ is the free-path integral against that weight, without any hard-core restriction.

Lemma 5.5 (Marked-path kernel). *For a fixed numerical $c_1 > 1$, which can be taken to be 3,*

$$\mathcal{K}_n(y, z) \leq l_n p_n(y) p_n(z) [e^{c_1 \delta \Delta_{O_n}}](y, z), \quad l_n = C \lambda (n+1) e^{C \lambda (n+1)} e^{-c n^2/t} \varepsilon^{1/3},\tag{5.10}$$

where Δ_{O_n} has Dirichlet boundary conditions, or periodic conditions if $O_n = \Lambda_K$. The three choices of E admit respectively

$$\varepsilon = 1, \quad \varepsilon = C e^{-c/t}, \quad \varepsilon = C_t D^{-12}.\tag{5.11}$$

The constants for the first two choices are uniform for $0 < t \leq 1$; the irregularity constant may depend on the fixed t and the previously chosen spatial parameters.

Proof. Fix a lift of y and one lift of z . Omit the marking factor temporarily but retain $J_A > 0$, E , and bin n . The resulting Euclidean kernel is bounded by each of

$$g_\delta(z - y), \quad C t^{-3/2} e^{-c n^2/t}, \quad C t^{-3/2} \varepsilon, \quad g_s(u) = (4\pi s)^{-3/2} e^{-|u|^2/(4s)}.\tag{5.12}$$

The first bound follows by discarding all tests. To prove the second, at least one half has at least $n/2$ completed exits when $n \geq 1$. Bound the opposite half's heat kernel by $Ct^{-3/2}$ and integrate the indicated half as ordinary Brownian motion; use reversal for the second half.

For completeness, let σ be a Brownian exit time from the unit ball. The maximal Gaussian bound gives $\mathbb{P}(\sigma \leq u) \leq Ce^{-c/u}$ for small u . Splitting the expectation at $u = s^{-1/2}$ therefore gives $\mathbb{E}e^{-s\sigma} \leq e^{-c'\sqrt{s}}$ for sufficiently large s . Successive exit durations are independent copies of σ , by the strong Markov property. If Q_t is their number completed by t , then

$$\mathbb{P}(Q_t \geq k) \leq e^{st}(\mathbb{E}e^{-s\sigma})^k.$$

Taking $\sqrt{s}$ a sufficiently small fixed multiple of k/t proves $\mathbb{P}(Q_t \geq k) \leq Ce^{-ck^2/t}$. When k/t is bounded the same statement follows by increasing C ; $n = 0$ is covered directly by the heat kernel supremum. This proves the second bound in Equation (5.12).

For the third bound, diameter greater than one forces one half to have diameter greater than $1/2$, since the two halves meet. Its Brownian probability is at most $Ce^{-c/t}$. Irregularity forces a failure on at least one half: an endpoint window, the mesh maximum, or the half-average with threshold C_{reg} . The proof of Lemma 5.3 bounds each such half failure by C_tD^{-12} . Once again the opposite half is bounded by its heat kernel supremum. This also proves the third bound for E_{all} with $\varepsilon = 1$.

Take the geometric mean of the three bounds. The surviving Gaussian is a constant multiple of $g_{3\delta}(z - y)$, and the other factors become $e^{-cn^2/t}\varepsilon^{1/3}$ after changing c, C . Equation (5.8) bounds the omitted mark by

$$e^{\lambda J_A} - 1 \leq C\lambda(n+1)e^{C\lambda(n+1)}.$$

For every lift with a nonzero contribution, its widened-time bridge has probability at least a fixed positive constant of staying in the unit tube about its interpolating segment. Indeed its centered bridge has duration $3\delta \leq 6$; Brownian scaling bounds this probability below by the positive small-ball probability at duration 6. The projected tube is contained in O_n , so its paths contribute to the killed kernel on O_n . Summing these bounds over the endpoint lifts is legitimate: the lifts give disjoint winding classes in the torus bridge decomposition. The cutoffs equal one at every contributing endpoint. This yields Equation (5.10). $\square$

The factor $\varepsilon^{1/3}$ records the rarity of the single-path failure. The confinement to O_n will ensure that the interacting moment costs a multiple of $|A|$, even when the number of marked paths is unrestricted.

5.4 Colors and the trace-sandwich estimate

Choose a constant m_0 sufficiently large, depending only on the upper increment constant in Proposition 3.5 and the fixed α -window. Define

$$M_n = (m_0 + n)/\delta, \quad \eta_n = l_n e^{m_0 + n}. \quad (5.13)$$

We choose m_0 so that, for $\delta \leq 2$,

$$M_n/2 \geq C_{\text{inc}}\alpha + 1, \quad \beta M_n/2 \geq 1, \quad (5.14)$$

where $C_{\text{inc}}\alpha$ bounds an upper particle-number increment and $D \geq 1$. The choice of m_0 is independent of t .

First truncate the permitted marked bins at $n \leq n_*$, so that the moment in Equation (5.7) becomes

$$\mathfrak{M}_{n_*} = \mathbb{E}_{\mathcal{R}} \prod_{i=1}^N \left[1 + (e^{\lambda J_A(\omega_i)} - 1) \mathbf{1}_E(\omega_i) \mathbf{1}_{\{n(\omega_i) \leq n_*\}} \right].$$

The enlarged one-particle space is

$$\mathfrak{h}_e = L^2(\Lambda_K) \oplus \bigoplus_{n=0}^{n_*} L^2(O_n).$$

Its first summand is called the original color, and the others are the marked colors. In each labelled color assignment of N particles impose the hard exclusion only between original-color particles. The direct sum of these allowed labelled spaces is denoted by $\mathcal{H}_e^{\text{lab}}$. On it let $\mathbb{H}^{\text{lab}}$ be the operator with the original-color hard-core kinetic form, and with one-particle operator $-c_1 \Delta_{O_n} + M_n$ on a particle of color n . There are no interactions involving a marked color. Simultaneous permutations of positions and colors commute with $\mathbb{H}^{\text{lab}}$. Denote their invariant subspace by $\mathcal{H}_e$, and denote the restriction of $\mathbb{H}^{\text{lab}}$ to it by $\mathbb{H}$.

On zero-extended original N -particle functions apply in each coordinate the map

$$v_e f = (f, (\sqrt{\eta_n} p_n f)_{0 \leq n \leq n_*}), \quad V_{\text{lab}} = v_e^{\otimes N}.$$

Here V_{lab} has domain $L^2(\Omega_N)$, the original labelled hard-core space, and takes values in $\mathcal{H}_e^{\text{lab}}$. It maps form-domain functions to form-domain functions: the original hard-core zeros remain, the marked cutoffs are compactly supported in their O_n , and all multipliers are Lipschitz. Define first the labelled operator

$$\mathcal{T}_{\text{lab}} = V_{\text{lab}}^* e^{-\delta \mathbb{H}^{\text{lab}}} V_{\text{lab}}.$$

The map V_{lab} intertwines coordinate permutations with simultaneous position-color permutations. Its restriction V_e to the original symmetric space therefore takes values in $\mathcal{H}_e$, and $\mathcal{T}_{\text{lab}}$ commutes with the original symmetrizer P_{sym} . Its symmetric restriction is

$$\mathcal{T} = \mathcal{T}_{\text{lab}}|_{\text{ran } P_{\text{sym}}} = V_e^* e^{-\delta \mathbb{H}} V_e.$$

Expanding the product defining the trajectory moment marks subsets of labels. For every marked subset, bin its paths and drop all collision prohibitions incident to them. The nonnegative path integral increases. Apply Lemma 5.5 to each marked path. Its coefficient is reproduced exactly by the color map, since $\eta_n e^{-\delta M_n} = l_n$. Each choice of marked labels and their bins appears once in the labelled color direct sum. Thus the tested labelled slab kernel is bounded pointwise by the kernel $K_{\mathcal{T}_{\text{lab}}}$ of $\mathcal{T}_{\text{lab}}$, before any restriction to symmetric functions.

The outer kernel $B_{\beta-\delta}^{\text{sym}}$ is nonnegative. Let H_N^{lab} denote the original labelled hard-core Hamiltonian. Integrating the pointwise bound against this outer kernel gives the symmetric trace by the following exact identity:

$$\begin{aligned} & \int B_{\beta-\delta}^{\text{sym}}(Z, Y) K_{\mathcal{T}_{\text{lab}}}(Y, Z) \, dY \, dZ \\ &= \text{Tr}_{\text{lab}} \left(e^{-(\beta-\delta) H_N^{\text{lab}}} P_{\text{sym}} \mathcal{T}_{\text{lab}} \right) = \text{Tr}_{\text{sym}} \left(e^{-(\beta-\delta) H_N} \mathcal{T} \right). \end{aligned}$$

Indeed both labelled operators commute with P_{sym} ; the product in the middle vanishes on its orthogonal complement and restricts to the final product on its range. The sole permutation normalization is the average $P_{\text{sym}} = N!^{-1} \sum_{\pi} U_{\pi}$ already present in the outer kernel. No factorial

is inserted in the labelled color expansion or in this trace identity. Consequently the truncated moment satisfies

$$\mathfrak{M}_{n_*} \leq \frac{\text{Tr}_{\text{sym}}(e^{-(\beta-\delta)H_N} \mathcal{T})}{Z_N} \leq \left(\frac{\text{Tr}_{\text{sym}} \mathcal{T}^{\beta/\delta}}{Z_N} \right)^{\delta/\beta}. \quad (5.15)$$

The second inequality is Schatten Hölder with exponents $\beta/(\beta-\delta)$ and β/δ . All operators have finite positive-time traces at the fixed finite volume and particle number.

To control the large power in Equation (5.15), we must account for the norm of the color map. Define

$$a(x) = \frac{1}{2} \log \left(1 + \sum_{n=0}^{n_*} \eta_n p_n(x)^2 \right), \quad A_0 = \sum_{i=1}^N a(x_i). \quad (5.16)$$

The same position multiplier acts in every color. The map $U_e = e^{-A_0} V_e$ is isometric, because the squared pointwise norm of its one-coordinate color vector is one. Therefore

$$\mathcal{T} = U_e^* X_0 U_e, \quad X_0 = e^{A_0} e^{-\delta \mathbb{H}} e^{A_0}.$$

Eigenvalue min–max for positive compact compressions gives $\text{Tr} \mathcal{T}^q \leq \text{Tr} X_0^q$ for every $q > 0$. The next lemma converts this sandwich into a heat trace with a bounded attractive field.

Lemma 5.6 (Trace bound for a Lipschitz sandwich). *For the preceding finite-color space and operators, set*

$$W_0 = 2a/\delta + c_1 |\nabla a|^2, \quad \widetilde{\mathbb{H}} = \mathbb{H} - \sum_{i=1}^N W_0(x_i).$$

For every real $q > 0$,

$$\text{Tr}_{\mathcal{H}_e} X_0^q \leq \text{Tr}_{\mathcal{H}_e} e^{-q\delta \widetilde{\mathbb{H}}}. \quad (5.17)$$

Proof. This is the sandwich estimate of OpenAI [35, Lemma “Trace bound for a Lipschitz sandwich”], with the hard-core Dirichlet conditions retained. We give the form argument, including the step that permits every positive power.

There are finitely many color assignments. In each, Dirichlet domain monotonicity bounds the heat trace by a free periodic or Dirichlet product heat trace. Hence $\mathbb{H}$ has compact resolvent and finite heat traces at positive times. Subtracting the bounded potential $\sum_i W_0(x_i)$ preserves these properties. The operator X_0 is positive, compact, and injective.

For $0 \leq s \leq \delta$ define

$$U(s) = e^{u(s)A_0} e^{-s\mathbb{H}} e^{A_0}, \quad u(s) = -1 + 2s/\delta.$$

Thus $U(0) = I$ and $U(\delta) = X_0$. Bounded Lipschitz multiplication preserves each form domain, including its original-color collision boundary and marked-color spatial boundary. This follows by applying the product rule to compactly supported approximants and taking their form-norm limits. For an initial vector f_0 and $s > 0$, spectral calculus puts $e^{-s\mathbb{H}} e^{A_0} f_0$ in the operator domain. The vector $f = U(s)f_0$ is norm differentiable and in the form domain. If $\mathfrak{q}$ is the form of $\mathbb{H}$, then

$$\frac{1}{2} \frac{d}{ds} \|f\|^2 = \frac{2}{\delta} \langle f, A_0 f \rangle - \text{Re } \mathfrak{q}(e^{u(s)A_0} f, e^{-u(s)A_0} f). \quad (5.18)$$

This identity only pairs a form-domain vector with the operator-domain semigroup vector; it does not require Lipschitz multiplication to preserve the operator domain.

In any color assignment, the kinetic coefficient c_i of coordinate i is 1 or c_1 . The product rule gives, almost everywhere,

$$\operatorname{Re} \left[\overline{\nabla_i(e^{uA_0} f)} \cdot \nabla_i(e^{-uA_0} f) \right] = |\nabla_i f|^2 - u^2 |\nabla a(x_i)|^2 |f|^2.$$

The cross terms have zero real part and the gap potentials commute with the multipliers. Since $|u| \leq 1$ and $c_i \leq c_1$, Equation (5.18) implies

$$\frac{1}{2} \frac{d}{ds} \|U(s)f_0\|^2 \leq -\tilde{\mathbf{q}}[U(s)f_0], \quad (5.19)$$

where $\tilde{\mathbf{q}}$ is the form of $\tilde{\mathbb{H}}$.

Let $\tilde{E}_1 \leq \tilde{E}_2 \leq \dots$ be the eigenvalues of $\tilde{\mathbb{H}}$ with multiplicity. Apply the same form calculation to $U(s)^{\otimes k}$, using the sums of A_0 and $\mathbb{H}$ over k copies of $\mathcal{H}_e$. Restrict these copies to their alternating subspace. In an eigenbasis of $\tilde{\mathbb{H}}$, the copy-sum Hamiltonian on that subspace has lower bound $\sum_{i=1}^k \tilde{E}_i$. The copies here are copies of the entire enlarged symmetric N -particle space; the particle symmetry within each copy is unchanged. Equation (5.19) consequently gives

$$\| \bigwedge^k U(\delta) \| \leq \exp \left(-\delta \sum_{i=1}^k \tilde{E}_i \right).$$

To justify integration at zero, integrate first from $s = \epsilon > 0$ and then let $\epsilon \downarrow 0$, using strong continuity and $U(0) = I$. The tensor form calculation is first valid on finite tensor sums and extends by the closed forms and density.

Write the decreasing eigenvalues of X_0 as $x_i > 0$. The norm of its k th exterior power is the product of its largest k eigenvalues. Hence

$$\sum_{i=1}^k \log x_i \leq -\delta \sum_{i=1}^k \tilde{E}_i \quad (k \geq 1).$$

For each real h , taking the supremum over $k \geq 0$ of the shifted partial sums proves

$$\sum_i (\log x_i - h)_+ \leq \sum_i (-\delta \tilde{E}_i - h)_+.$$

Both sums have finitely many positive terms for fixed h , by compactness. For every $q > 0$,

$$e^{qu} = \int_{\mathbb{R}} q^2 e^{qh} (u - h)_+ \, dh.$$

Integrate the preceding positive-part inequality and apply Tonelli's theorem. It gives $\sum_i x_i^q \leq \sum_i e^{-q\delta \tilde{E}_i}$, which is Equation (5.17). $\square$

5.5 The interacting trajectory moment

The remaining task is to compare the enlarged heat trace with Z_N . The original color will be estimated by the field-density bound; the positive gaps will pay for the other colors. Spatial confinement in Equation (5.9) keeps both costs proportional to the number of tested sites.

Proposition 5.7 (Joint trajectory moment). *Use any of the three events E and the coefficients l_n, η_n, M_n from Equations (5.10) and (5.13). For a sufficiently large fixed constant C_0 , assume*

$$\frac{C_0}{\delta} \sum_{n \geq 0} \eta_n \leq 1. \quad (5.20)$$

Then for every nonempty set A of sites,

$$\begin{aligned} & \log \mathbb{E}_{\mathcal{R}} \exp \left(\lambda \sum_{i=1}^N J_A(\omega_i) \mathbf{1}_E(\omega_i) \right) \\ & \leq CD|A| \sum_{n \geq 0} (n+1)^3 \eta_n + C\delta T|A|(V^{-1} + T^{3/2}) \sum_{n \geq 0} (n+1)^3 e^{-\beta M_n/2}. \end{aligned} \quad (5.21)$$

All constants are independent of n_, K, D, T and $0 < t \leq 1$, with the stated t -dependence retained in ε for irregularity.*

Proof. For the truncated normalization in Equation (5.16), write $S(x) = \sum_{n \leq n_*} \eta_n p_n(x)^2$. Cauchy–Schwarz gives

$$|\nabla a|^2 \leq \frac{S}{(1+S)^2} \sum_{n \leq n_*} \eta_n |\nabla p_n|^2 \leq C \sum_{n \leq n_*} \eta_n \mathbf{1}_{O_n}.$$

Also $2a \leq S$. Since $\delta \leq 2$,

$$0 \leq W_0 \leq \frac{C}{\delta} \sum_{n \leq n_*} \eta_n \mathbf{1}_{O_n} \leq 1, \quad (5.22)$$

where the last inequality follows from Equation (5.20).

For every $0 \leq J \leq N$, differentiation with respect to a bounded field strength gives

$$\frac{d}{du} \log Z_J^{(-uW_0)} = \beta \mathbb{E}_{J, -uW_0} \sum_{i=1}^J W_0(x_i) \quad (0 \leq u \leq 1).$$

At finite volume this identity follows from the Duhamel formula and trace cyclicity. The field is bounded and the positive-time traces are finite. Apply Corollary 4.4 to the cell covers in Equation (5.9), and then integrate in u . It follows that

$$\log \frac{Z_J^{(-W_0)}}{Z_J} \leq \frac{\beta C}{\delta} D|A| \sum_{n \leq n_*} (n+1)^3 \eta_n. \quad (5.23)$$

The vacuum case has zero left side. In addition, Proposition 3.5 gives

$$\frac{Z_J}{Z_N} \leq e^{\beta C_{\text{inc}} \alpha (N-J)}. \quad (5.24)$$

Evaluate Lemma 5.6 at $q = \beta/\delta$. The enlarged symmetric space decomposes by the number J of original-color particles and the occupation numbers k_n of each marked color, with $J + \sum_n k_n = N$. Each sector is one normalized tensor product of the symmetric species spaces. There is no multinomial multiplicity: symmetrizing the mutually orthogonal assignments of colors gives an isometry from this tensor product onto the sector. Its original factor has trace $Z_J^{(-W_0)}$; the marked factors are ideal Bose canonical traces on O_n .

Use Equations (5.23) and (5.24). Absorbing the latter factor into the marked energies subtracts $C_{\text{inc}} \alpha$ per marked particle. With $W_0 \leq 1$ and Equation (5.14), each such one-particle operator is

bounded below in form by $-c_1\Delta_{O_n} + M_n/2$. The variational eigenvalue principle therefore bounds every fixed marked canonical trace by that operator's trace. We may then drop the restriction $\sum_n k_n \leq N$ and sum the independent marked occupations. The original count is determined by those occupations before this restriction is dropped, so no extra sum over J is introduced.

The logarithm of the ideal grand trace for color n is

$$\sum_{a \geq 1} \frac{e^{-a\beta M_n/2}}{a} \text{Tr} e^{a\beta c_1 \Delta_{O_n}} \leq C|O_n|(V^{-1} + T^{3/2})e^{-\beta M_n/2}.$$

Here the periodic Gaussian sum and killing give $\text{Tr} e^{sc_1 \Delta_{O_n}} \leq C|O_n|(V^{-1} + s^{-3/2})$, and $\beta M_n/2 \geq 1$ controls the sum in a . Combining this with the two preceding comparisons proves

$$\begin{aligned} \log \frac{\text{Tr} e^{-\beta \tilde{\mathbb{H}}}}{Z_N} &\leq \frac{\beta C}{\delta} D|A| \sum_{n \leq n_*} (n+1)^3 \eta_n \\ &\quad + C|A|(V^{-1} + T^{3/2}) \sum_{n \leq n_*} (n+1)^3 e^{-\beta M_n/2}. \end{aligned}$$

Multiply by $\delta/\beta = \delta T$ and use Equation (5.15), the compression inequality, and Lemma 5.6. This proves Equation (5.21) for the truncated product.

For each path the truncated marking weight increases to the full marking weight as $n_* \rightarrow \infty$. The finite product over labels also increases. Monotone convergence removes the truncation, while the displayed bounds on the right are uniform in n_* and summable. This proves the proposition. $\square$

5.6 Completion of the certificate estimate

We first record the sizes of the two costs in Equation (5.21). The Gaussian bin tail in Equation (5.10) implies

$$\sum_{n \geq 0} (n+1)^3 \eta_n \leq C_\lambda \lambda \varepsilon^{1/3}, \quad (5.25)$$

where C_λ is bounded on bounded λ -intervals, uniformly for $0 < t \leq 1$. Indeed the summand is bounded by a constant times $\lambda \varepsilon^{1/3} (n+1)^4 e^{C\lambda(n+1)+n-cn^2}$, including the fixed factor e^{m_0} . The same estimate without $(n+1)^3$ controls the smallness condition. Since $\beta \geq D$ and $\delta \leq 2$, the second term of Equation (5.21) is $o(1)|A|$ as $D \rightarrow \infty$, uniformly in $0 < t \leq 1$ and $K \geq 100$.

Proof of Proposition 5.4. It suffices to prove arbitrarily small exponential bases for simultaneous failures of each of the six individual tests. At both endpoints the law is μ , by Proposition 5.1. Thus the low-count test at Y and the close-count tests at Y, Z have this property by Propositions 4.5 and 4.6.

For the number of visitors choose $E = E_{\text{all}}$ and $\lambda = c't$, where $c' > 0$ is sufficiently small and fixed. Equation (5.25) makes Equation (5.20) hold uniformly for $0 < t \leq 1$, and the moment's first coefficient is at most $C\lambda D$ with C independent of t . If every site of A has more than $H_1 D$ visitors, then $\sum_i J_A(\omega_i) > H_1 D|A|$. Exponential Markov inequality yields

$$\mathcal{R}(\text{this event}) \leq \exp\{[-\lambda(H_1 - C)D + o(1)]|A|\}.$$

Choose $H_1 > C + 1$ now. For every subsequently fixed $t > 0$, the exponential base tends to zero as D increases. This choice of H_1 is unaffected by reducing t .

For large-diameter visitors take $\lambda = 1$ and $E = E_{\text{diam}}$. Now $\varepsilon^{1/3} \leq C e^{-c/t}$. Choose t sufficiently small that both Equation (5.20) and the bound $CD \sum_n (n+1)^3 \eta_n \leq \theta D/2$ hold. If every site of

A has at least θD such visitors, their total number of visits to the tested balls is at least $\theta D|A|$. Equation (5.21) therefore bounds this probability by $\exp[-(\theta D/2 - o(1))|A|]$.

Finally fix a large $\lambda > 0$ and take $E = E_{\text{irr}}$. With t now fixed, Equation (5.25) gives $\sum_n (n+1)^3 \eta_n = O_{\lambda,t}(D^{-4})$. The smallness condition holds for large D , and the whole coefficient per site in the moment tends to zero. If each tested site has even one irregular visitor, the sum of its irregular visits is at least $|A|$, counting a path once for every tested ball it visits. Its probability is consequently at most $\exp[-(\lambda - o(1))|A|]$. Choosing λ large and then D large makes this base as small as prescribed.

For clarity, no independence between tests or between cells is needed to combine them. Given a failure at every site, assign one of the six failure types to each site. There are at most $6^{|A|}$ assignments. In each, some type occurs on a prescribed subset of size at least $|A|/6$. If every simultaneous single-type failure has probability at most $(p')^{|A|}$, with $0 < p' < 1$, the union is bounded by $(6(p')^{1/6})^{|A|}$. Take p' so small that $6(p')^{1/6} \leq p_1$. The preceding choices and a final increase of D_0 ensure all six estimates. The empty set satisfies Equation (5.6) trivially. All arguments survive a further reduction of the fixed t , followed by the corresponding increase of D_0 . $\square$

We have obtained an unconditional joint bound for failure of the full-path certificates, with no loss proportional to the total volume. The next two sections use these certificates to find movable paths and then expose their environment while preserving an exact conditional bridge law.

6 Moving a path through a neighboring cell

The certificates from Section 5 control how many paths can obstruct a local move. We now show that most particles in a certified cell admit many replacements through each neighboring cell. The argument is deterministic in the obstacles and their endpoints; the thermal law enters only through the certificate supply.

We reproduce the bridge estimates of OpenAI [36, Section 5]. They separate two different averaging operations. Averaging over many separated endpoint pairs controls collisions against order D obstacles. For one endpoint pair, averaging over the proposed midpoint makes the collision probability against a single regular obstacle tend to zero. The distinction will matter when two replacement paths share a midpoint.

We use the regularity conditions of Definition 5.2. Thus $r = \alpha/D$, $s_0 = bD^{-1/3}$, $h_0 = D^{-1}$, and $\gamma = 1/12$. Each half of $[0, 2t]$ has $m_t = \lceil t/r^2 \rceil$ mesh intervals of length $\tau = t/m_t$. For a regular path c , its lifted diameter on interval k is $rA_k(c)$, where

$$\max_k A_k(c) \leq D^\gamma, \quad \sum_{k=1}^{2m_t} (1 + A_k(c))^3 \leq 2C_{\text{reg}} m_t. \quad (6.1)$$

Its diameter in each outer time window of length h_0 is at most $s_0/8$. All estimates below hold for each fixed $0 < t \leq 1$ and sufficiently large D , uniformly in $\alpha \in [\alpha_0/2, 2\alpha_0]$. In particular, we may require

$$r^2/2 \leq \tau \leq h_0/2, \quad h_0 < t/4, \quad rD^\gamma < 1, \quad C\tau/t \leq r \quad (6.2)$$

for any fixed geometric constant C used below, and also $r < 3s_0/4$.

Fix a cell $B_v = v + [-1/2, 1/2]^3$. Suppose that a path's outer endpoints y, z have local lifts satisfying $y \in B_v$ and $|z - v| \leq 3$. A *target cell* is B_v or one of its six face-neighbor cells. For u in a target cell B , let $Q_{y,z}^u$ be the law obtained by joining two independent Euclidean Brownian bridges,

of generator Δ , from y to u and from u to z , each of duration t . We project this path to the torus. Let

$$Q_{y,z}^B = \int_B Q_{y,z}^u du, \quad Q_{y,z}^0 = \text{the Euclidean bridge from } y \text{ to } z \text{ of duration } 2t.$$

Both laws are projected to the torus. Unit cells have volume one, so $Q_{y,z}^B$ is a probability measure. Write $\mathcal{C}_v$ for the event that the replacement is regular and its local lift stays in the closed ball of radius 10 about v . For a deterministic torus path c , define its collision event by

$$\mathcal{H}(c) = \{\omega : d(\omega(s), c(s)) \leq r \text{ for some } s \in [0, 2t]\}.$$

The estimates concern $\mathcal{C}_v \cap \mathcal{H}(c)$ under the unconditioned bridge laws. No independence between the cutoff event and the collision event is assumed.

6.1 Separated endpoints and Gaussian densities

Lemma 6.1 (Averaged Gaussian density). *Let $y_1, \dots, y_m \in \mathbb{R}^3$ have pairwise distances at least s_0 , where $m \geq \ell D/2$ and $s_0 = bD^{-1/3}$. Fix $u \in \mathbb{R}^3$ and $0 < s \leq t/2$. The average of the time- s densities of duration- t Brownian bridges from y_i to u is bounded everywhere by*

$$C \left(1 + \frac{1}{Ds^{3/2}} \right). \quad (6.3)$$

Here C depends on b, ℓ , but not on t, s, D or u .

Proof. The means are $ay_i + (1-a)u$, where $a = 1 - s/t \geq 1/2$, and the covariance matrix is $2s(1 - s/t)$ times the identity. The means are therefore $s_0/2$ -separated. For points q_i separated by $s_0/2$, disjoint balls of radius $s_0/4$ give

$$\#\{i : |q_i - x| \leq R\} \leq C(1 + R/s_0)^3.$$

Split the Gaussian sum into the ball of radius $\sqrt{s}$ about x and the annuli with radii $j\sqrt{s}, (j+1)\sqrt{s}$, $j \geq 1$. The last bound, multiplied by the Gaussian decay on each annulus, yields

$$\sum_i \exp\left(-\frac{|x - q_i|^2}{4s}\right) \leq C \left(1 + \frac{s^{3/2}}{s_0^3} \right).$$

The covariance lies between s and $2s$ times the identity, so the averaged density is at most

$$\frac{C}{ms^{3/2}} + \frac{C}{ms_0^3} \leq C \left(\frac{1}{Ds^{3/2}} + 1 \right).$$

□

6.2 Collision with a deterministic obstacle

Lemma 6.2 (Averaged collision bound). *Fix $m \geq \ell D/2$ pairs (y_i, z_i) with $y_i \in B_v$ and $|z_i - v| \leq 3$. Assume that each of the two endpoint lists is s_0 -separated. Let c be a deterministic regular torus path whose outer endpoints are at distance greater than s_0 from the corresponding endpoints of every label under consideration. Then, for every target cell B ,*

$$\frac{1}{m} \sum_{i=1}^m Q_{y_i, z_i}^B(\mathcal{C}_v \cap \mathcal{H}(c)) \leq Ctr + C \frac{rD^{3\gamma}}{D\sqrt{h_0}}. \quad (6.4)$$

The same conclusion holds if c is the original path of one of these labels: omit that label's summand and retain the divisor m , assuming endpoint separation for all the remaining labels. The constant C may depend on b, ℓ, C_{reg} , but is independent of small t .

Proof. We estimate the first leg from an outer endpoint y_i to the random midpoint u ; reversing time gives the identical estimate for the leg from z_i to u . On $\mathcal{C}_v$, the endpoint windows of both paths have diameter at most $s_0/8$. Endpoint separation consequently leaves distance at least $3s_0/4 > r$ throughout $[0, h_0]$. Every mesh interval on which a collision remains possible starts at $s \geq h_0 - \tau \geq h_0/2$.

We first explain the use of lifts. On a mesh interval, the obstacle has diameter at most $rD^\gamma < 1$. A collision with a confined replacement therefore requires $d(c(s), v) < 12$ at the interval's starting time. For $K > 50$ this determines a unique lift $\tilde{c}(s)$ within distance 12 of the chosen lift of v ; continue that lift over the interval. Its displacement is at most $rA_k(c)$. Intervals for which no such lift exists make no contribution. We may thus use Euclidean Gaussian densities without summing over windings.

The first half of a leg. Suppose $s \leq t/2$ and condition on u . If X_s is the position of the trial bridge at time s , its segment of length τ has the representation

$$X_{s+q} = X_s + \frac{q}{t-s}(u - X_s) + R_q, \quad 0 \leq q \leq \tau. \quad (6.5)$$

Here R is the initial segment of a centered bridge of duration $t-s$, independent of X_s . Its law does not depend on u or on the endpoint label. This is the Gaussian conditional law of a bridge after its time- s position has been specified. Equivalently, $R_q = W_q - qW_{t-s}/(t-s)$ for a Brownian motion W of generator Δ . Gaussian maximal estimates, or the Brownian reflection bound integrated against $3a^2 da$, give

$$B_s = \frac{1}{r} \sup_{0 \leq q \leq \tau} |R_q|, \quad \mathbb{E}B_s^3 \leq C, \quad (6.6)$$

because $\tau \leq r^2$ and $\tau \leq t-s$.

On the cutoff event, $|X_s - v| \leq 10$, whereas every target point is within distance 2 of v . The drift term in (6.5) is thus at most $C\tau/t \leq r$. A collision on this interval, together with the cutoff, implies

$$|X_s - \tilde{c}(s)| \leq r(2 + A_k(c) + B_s). \quad (6.7)$$

We use the cutoff only to obtain this necessary event. We now discard it and integrate the unrestricted bridge law. By Lemma 6.1, the density of X_s , averaged over the labels, is bounded by (6.3). Since B_s is independent of X_s and has the common bound (6.6), the averaged probability of (6.7) is at most

$$Cr^3(1 + A_k(c))^3 \left(1 + \frac{1}{Ds^{3/2}}\right). \quad (6.8)$$

This estimate holds for each u , hence also after averaging u over the target cell. Omitting one label only decreases the unnormalized Gaussian sum and leaves the same bound with divisor m .

The second half of a leg. Suppose $s > t/2$. For a fixed label, write the whole unrestricted leg as

$$X_q = (1 - q/t)y_i + (q/t)U + \beta_q, \quad 0 \leq q \leq t,$$

where U is uniform in B and the centered bridge β is independent of U . Condition on the entire path β . The conditional density of X_s is at most $(t/s)^3 \leq 8$. The mean displacement during the mesh interval is at most $C\tau/t \leq r$, uniformly in $U \in B$. Put

$$\tilde{B}_s = r^{-1} \sup_{0 \leq q \leq \tau} |\beta_{s+q} - \beta_s|.$$

Writing $\beta_q = W_q - qW_t/t$ gives $\mathbb{E}\tilde{B}_s^3 \leq C$: the Brownian increment contributes $C\tau^{3/2}/r^3$, and the linear correction contributes $C\tau^3/(r^3t^{3/2})$, both bounded. Conditional on β , a collision requires the ball in (6.7), with B_s replaced by $\tilde{B}_s$. Both its radius and center are then fixed while U retains its uniform distribution. Integrating first over U and then over β bounds the probability by

$$Cr^3(1 + A_k(c))^3. \quad (6.9)$$

In particular, no independence between X_s and the fluctuation increment is needed here.

Summing the intervals. Apply the two bounds to both legs. The terms without $s^{-3/2}/D$ use the averaged regularity condition:

$$Cr^3 \sum_{k=1}^{2m_t} (1 + A_k(c))^3 \leq Cr^3 m_t \leq Ctr.$$

Here $m_t = t/\tau$ and $\tau \geq r^2/2$. For the remaining terms use $(1 + A_k(c))^3 \leq CD^{3\gamma}$ and

$$\sum_{\substack{s=j\tau \geq h_0/2 \\ s \leq t/2}} s^{-3/2} \leq \frac{C}{\tau\sqrt{h_0}} \leq \frac{C}{r^2\sqrt{h_0}}.$$

Their contribution is therefore at most $Cr^3 D^{3\gamma}/(Dr^2\sqrt{h_0}) = CrD^{3\gamma}/(D\sqrt{h_0})$. These two sums give (6.4). Every constant in these sums is independent of t ; making t smaller only increases the required lower threshold on D in (6.2). $\square$

Lemma 6.3 (A single replacement). *Let $y \in B_v$, $|z - v| \leq 3$, and let c be a deterministic regular path satisfying $d(y, c(0)) > s_0$ and $d(z, c(2t)) > s_0$. For either $Q = Q_{y,z}^B$, with any target cell B , or $Q = Q_{y,z}^0$, one has*

$$Q(\mathcal{C}_v \cap \mathcal{H}(c)) \leq Ctr + C \frac{rD^{3\gamma}}{\sqrt{h_0}} \leq C\alpha(tD^{-1} + D^{-1/4}). \quad (6.10)$$

Thus this probability tends to zero with D for each fixed t .

Proof. For $Q_{y,z}^B$, repeat the preceding proof with one endpoint. At $s \leq t/2$ its Gaussian density is at most $Cs^{-3/2}$; the bound (6.8) becomes $Cr^3(1 + A_k(c))^3 s^{-3/2}$. The second half of each leg retains (6.9). The same two sums give the first inequality.

For $Q_{y,z}^0$, use the forward orientation up to time t and the reversed orientation for the rest. In either orientation the remaining bridge duration is at least t . Formula (6.5) holds with $2t$ in place of t , its drift on the cutoff event is at most $C\tau/t$, and its time- s density is bounded by $Cs^{-3/2}$ for $0 < s \leq t$. Endpoint separation again excludes the window $[0, h_0]$. Consequently the first-half argument and its mesh sum apply over both orientations, giving $CrD^{3\gamma}/\sqrt{h_0}$. Adding the nonnegative term Ctr yields the stated common bound. Finally $r = \alpha/D$, $h_0 = D^{-1}$ and $3\gamma = 1/4$ give the last inequality. $\square$

Remark 6.4. The midpoint in $Q_{y,z}^B$ is averaged over a unit cell. An estimate uniform in a prescribed midpoint is false: if $u = c(t)$, the trial collides at time t . Restricting a uniform midpoint to a measurable set of volume at least $a > 0$ multiplies an upper bound for a nonnegative event by at most a^{-1} . Conditioning a trial on an event of probability at least a has the same cost. These elementary changes of measure will allow us to use Lemma 6.3 after choosing successful trials.

6.3 Many paths can be moved successfully

The averaged estimate now pays for all nearby obstacles. The following statement separates that deterministic estimate from the random cell certificates that will supply its hypotheses.

Corollary 6.5 (Supply of successful labels). *Consider a collection of deterministic torus paths with endpoints (y_j, z_j) , and fix a cell v . Suppose that at most $H_1 D$ of these paths visit the ball of radius 12 about v , and that every visiting path is regular. Suppose a list of $m \geq \ell D/2$ labels satisfies $y_i \in B_v$, $|z_i - v| \leq 3$, and that both endpoints of every listed label are at distance greater than s_0 from every other endpoint in the corresponding full configuration.*

Call a trial for label i successful if it satisfies C_v and avoids every obstacle path except the original path of i . For sufficiently small fixed $t > 0$ and then sufficiently large D , at least $9m/10$ listed labels have the following property: for each of the seven target cells B , the set

$$T_{i,B} = \{u \in B : Q_{y_i, z_i}^u(\text{successful trial}) \geq 9/10\} \quad (6.11)$$

has volume at least $99/100$. All choices are uniform in the deterministic collection, v , and $\alpha \in [\alpha_0/2, 2\alpha_0]$.

Proof. The linear means of both bridge legs stay within distance 3 of v . The Gaussian maximal bound therefore makes confinement failure at most $Ce^{-c/t}$. Lemma 5.3 gives a regularity failure probability $\varepsilon_D(t) \rightarrow 0$, uniformly over the listed endpoints and the target points. A confined trial can collide only with an obstacle visiting the ball of radius 12, since $r < 1$. Union-bounding over those obstacles and using Lemma 6.2, with a self-label summand omitted when necessary, gives for each target cell

$$\begin{aligned} \frac{1}{m} \sum_i \int_B Q_{y_i, z_i}^u(\text{failure}) du &\leq Ce^{-c/t} + \varepsilon_D(t) + CH_1 D \left(tr + \frac{rD^{3\gamma}}{D\sqrt{h_0}} \right) \\ &\leq Ce^{-c/t} + \varepsilon_D(t) + CH_1 \alpha(t + D^{-1/4}). \end{aligned} \quad (6.12)$$

The constant multiplying t is independent of small t , which is the reason for the two parts of the bridge estimate. First choose t small and then D large so that the last bound is less than $1/70000$ for every α in the fixed window.

If $|T_{i,B}| < 99/100$, the integral of the failure probability over B is greater than $(1/100)(1/10) = 1/1000$. Thus fewer than $m/70$ labels fail the conclusion for a specified target cell. There are seven target cells, so fewer than $m/10$ labels fail for at least one of them. All the remaining labels have the asserted property. $\square$

7 Exposing the environment and retaining movable groups

We now retain a small group of particles at each cell and fix the rest of the path configuration. For the retained paths we need two facts: most cells contain many labels that can be moved, and their exact conditional law is a product of one-path laws conditioned on mutual avoidance. The first assertion is averaged over the environment; the second holds at almost every fixed environment.

The selection and exposure argument is that of OpenAI [36, Section 6]. Here the exposed environment also contains W' , the additional configuration in the thermal purification. Its weight depends only on the outer endpoints, so the conditional bridge argument remains valid.

7.1 Eligible labels and their local trial laws

The endpoints Y, Z and path regularity are those of Section 5. A label i is *eligible at v* if

- (i) $Y_i \in B_v$ and $d(Z_i, v) \leq 3$;
- (ii) $d(Y_i, Y_j) > s_0$ and $d(Z_i, Z_j) > s_0$ for every $j \neq i$;
- (iii) its trajectory is regular and remains in the radius-10 ball about v .

The eligible sets at distinct sites are disjoint, since the starting cells form a partition. When K is large, the confinement ball lifts injectively to Euclidean space, so the endpoints have unique lifts near the chosen lift of v .

For each eligible label i at v , a *target cell* is B_v or one of its six face-neighbor cells. Write Q_i^u for the probability law of two independent Euclidean bridges, each of duration t , from Y_i to u and from u to Z_i , projected to the torus. All lifts in this definition are near v . For a collection F of fixed obstacle paths with distinct labels other than i , put

$$C_i(F) = \{\text{the trial is regular and stays within distance 10 of } v; \\ \text{it avoids every path in } F \text{ at distance } r\}. \quad (7.1)$$

Avoidance throughout means strict distance greater than r at every time. The endpoints are fixed in this notation.

Lemma 7.1 (Useful labels in a certified cell). *At a cell with a certificate there are at least $\ell D/2$ eligible labels. After t has been chosen sufficiently small and D sufficiently large, at least nine tenths of these labels satisfy the following stronger property: for every target cell B_w ,*

$$\left| \left\{ u \in B_w : Q_i^u(C_i(F_i^{\text{all}})) \geq 0.9 \right\} \right| \geq 0.99. \quad (7.2)$$

Here F_i^{all} consists of all other trajectories in the sampled configuration, and $|\cdot|$ denotes spatial volume.

Proof. Start with the at least ℓD labels having $Y_i \in B_v$. Every such path visits the radius-12 ball and is therefore regular. Discard the fewer than θD visiting paths of diameter greater than one. The remaining paths stay within distance 2 of v and end within distance 2 of v . Discard also those whose initial or terminal endpoint has another endpoint within distance s_0 . The certificate bounds these two losses by θD each, since the endpoints in question lie in the radius-5 ball about v . The remaining labels are eligible, and their number is at least $(\ell - 3\theta)D \geq \ell D/2$.

We now apply Corollary 6.5 to the *entire* eligible list at this site, not merely the subset just used to prove its lower cardinality bound. Every eligible label has its initial endpoint in B_v , terminal endpoint within distance three of v , and both endpoints separated from all corresponding endpoints of other labels by more than s_0 . The certificate supplies at most $H_1 D$ visiting obstacles, all regular. These are precisely the hypotheses of that corollary. It gives Equation (7.2) for at least nine tenths of the entire eligible list, with choices uniform in the certificate. $\square$

7.2 Disjoint groups and the information retained

Choose an integer $M \geq 1$, to be fixed after the one-path comparison constant below, and put

$$G_D = \left\lfloor \frac{\ell D}{4M} \right\rfloor. \quad (7.3)$$

We take D large enough that $G_D \geq 1$. At a site with at least MG_D eligible labels, choose an ordered list of MG_D distinct eligible labels uniformly without replacement, and divide it into G_D successive groups of size M . If there are fewer, all groups at that site are empty. The allocation randomizations are independent between sites conditional on the full path sample, and use the eligible lists only.

Fix one group index $g \in \{1, \dots, G_D\}$. Let $\mathcal{I}$ be the union of its labels over all sites. The *exposed datum* $\mathcal{B}$ comprises

- both endpoint configurations, W' , every eligibility indicator, and all allocation information;
- the complete trajectories of labels outside $\mathcal{I}$.

These last trajectories form the frozen family F . The law of $\mathcal{B}$ is the pushforward of the thermal slab law $\mathcal{R}$ and the allocation randomizations. No certificate event is included in this datum. Subsequent expectations over $\mathcal{B}$ always use this unconditional law.

For $i \in \mathcal{I}$ at site v , let B_i be the unconditioned Euclidean bridge law of duration $2t$ between its near endpoint lifts. Define

$$P_i(d\omega_i) = \frac{\mathbf{1}_{C_i(F)} B_i(d\omega_i)}{a_i}, \quad a_i = B_i(C_i(F)). \quad (7.4)$$

The event $C_i(F)$ includes the individual regularity and confinement cutoffs, as in Equation (7.1).

Lemma 7.2 (Exact conditional law). *For almost every exposed datum the normalizers in Equation (7.4) are positive, and the conditional law of the unexposed trajectories is*

$$\lambda_{\mathcal{I}}(d\omega_{\mathcal{I}}) = \frac{1}{Z_{\mathcal{I}}} \mathbf{1}_{\{\text{all pairs in } \mathcal{I} \text{ avoid each other}\}} \prod_{i \in \mathcal{I}} P_i(d\omega_i), \quad Z_{\mathcal{I}} > 0. \quad (7.5)$$

All reference trajectories obey their own regularity and confinement cutoffs. Each label has at most CM other labels in this group with which a collision can be possible, where C is a fixed geometric constant.

Proof. Condition first on Y, Z, W' . In the slab law of Proposition 5.1, the two outer heat kernels are then constant. The remaining free reference is a product of torus bridges, weighted only by the indicator of mutual hard avoidance. At fixed endpoints, eligibility of an individual path consists of its own regularity and confinement conditions: its starting cell, endpoint displacement and endpoint separation are already fixed. The probabilities of all allocations depend only on the recorded eligibility indicators, and are consequently constant on the conditional fiber.

Now expose all paths outside $\mathcal{I}$. Every retained label was eligible, so the only remaining individual restrictions are exactly its two cutoffs and avoidance of the fixed frozen paths. Confinement to its radius-10 ball selects a single Euclidean lift when K is large. The winding weights of the original torus bridge then depend only on the fixed endpoints and cancel in normalization. The remaining nonfactorizing restriction is mutual avoidance inside $\mathcal{I}$. This proves Equation (7.5) by disintegration on the continuous-path spaces. Equivalently, integrate the asserted formula against bounded functions of the exposed and unexposed variables and use Fubini; all factors just identified reproduce the thermal slab density and the allocation probabilities.

The integral of the conditional density is positive for almost every sampled datum. Its positivity implies positivity of each a_i and of $Z_{\mathcal{I}}$; otherwise that fiber would have zero weight. Finally a retained trajectory stays within distance ten of its home site. Two such tubes can meet at distance $r < 1$ only when their home sites are within a fixed lattice distance. There are at most M labels per site and a fixed number of such sites, proving the last assertion. In particular Equation (7.5) is a conditioned product, not a claim that its trajectories are independent. $\square$

7.3 Accessibility and the probability of bad cells

For $i \in \mathcal{I}$ and one of its target cells B_w , set

$$T_i(w) = \{u \in B_w : Q_i^u(C_i(F)) \geq 0.9\}. \quad (7.6)$$

A label is *accessible* if $|T_i(w)| \geq 0.99$ for all seven target cells. This property and all the sets $T_i(w)$ depend only on $\mathcal{B}$. Let $\mathcal{A}_v$ be the accessible labels of the fixed group at v . A site is *good* if $|\mathcal{A}_v| \geq M/2$, and *bad* otherwise. The sets in Equation (7.6) are measurable: bridge kernels are measurable in their endpoints, and regularity, confinement and strict avoidance are Borel events of continuous paths.

Proposition 7.3 (Unconditional supply of good cells). *For any prescribed $p > 0$, the constants before M can be chosen as above, and then M can be chosen sufficiently large, so that for sufficiently large D and K ,*

$$\mathbb{P}_{\mathcal{B}}\{A \subset \{\text{bad sites}\}\} \leq p^{|A|} \quad (7.7)$$

for every set A of distinct sites. The bound is uniform in the group index, $0 < T \leq D^{-1}$, and the allowed α -window. It remains valid if M is increased before increasing the lower threshold on D .

Proof. Choose $p_1 < p/2$ in Proposition 5.4 and meet also the requirements of Lemma 7.1. Fix the full path sample before the allocation randomizations. At a certified cell there are at least $\ell D/2 \geq MG_D$ eligible labels, and at least nine tenths satisfy Equation (7.2). For a fixed group, its M labels form a uniform sample without replacement from this list. For any k specified positions in that sample, the probability that all k lack the stronger property is at most 10^{-k} . A union bound therefore gives

$$\mathbb{P}\{\text{fewer than } M/2 \text{ stronger labels in the group} \mid \text{full sample}\} \leq q_M := 2^M 10^{-M/2}. \quad (7.8)$$

This tends to zero exponentially in M .

A label with the stronger property is accessible after allocation: its frozen obstacle family is a subset of all other sampled paths, so removing obstacles can only increase its success probabilities. Moreover the auxiliary sampling tests at distinct certified cells are independent conditional on the full sample. For each subset $A_0 \subset A$, designate the cells of A_0 as lacking certificates and the other cells as certified but failing the sampling test. The probability of this event is at most $p_1^{|A_0|} q_M^{|A \setminus A_0|}$, by conditioning first on the full sample and then using Proposition 5.4. Summing over A_0 bounds the probability in Equation (7.7) by $(p_1 + q_M)^{|A|}$. Take M large enough that $q_M < p/2$. The argument applies to each group under the unconditional law of the exposed datum. $\square$

7.4 One-path comparisons at fixed exposed data

The remaining estimates apply at a fixed datum, even when its unexposed trajectories no longer satisfy any full-sample certificate. Let i be accessible, let B_w be a target cell, and let $U \subset T_i(w)$ be a measurable set of volume at least 0.98. Define the proposal Π_i^U by choosing u uniformly in U and then sampling Q_i^u conditioned on $C_i(F)$. The target set is chosen using exposed data only; it must not depend on the trajectories of other unexposed labels.

Proposition 7.4 (Density and deterministic compatibility). *There is a constant $G \geq 2$, fixed after t and independent of D, K, M , such that*

$$\Pi_i^U \ll P_i, \quad 0 \leq \frac{d\Pi_i^U}{dP_i} \leq G, \quad \int \frac{d\Pi_i^U}{dP_i} dP_i = 1. \quad (7.9)$$

There is also $e_D \rightarrow 0$, uniform in the exposed datum, with the following property. Fix any regular, confined trajectory of a distinct label in $\mathcal{I}$, keeping its recorded endpoints. Under either P_i or Π_i^U , the probability of colliding with that deterministic trajectory is at most e_D . Its label need not be accessible. One may take

$$e_D = C_t(tr + rD^{3\gamma}h_0^{-1/2}) \leq C'_t D^{-1/4}. \quad (7.10)$$

Proof. The lifted midpoint density g_i of B_i is the Euclidean Gaussian with mean $(Y_i + Z_i)/2$ and covariance tI . On every target cell it has a lower bound $c_t > 0$, uniform in the eligible endpoint data. Indeed those endpoints are within distance three of v , and every target cell lies in a fixed bounded neighborhood of v ; the explicit Gaussian density gives such a bound. Disintegrating B_i at its midpoint gives the conditional laws Q_i^u . Hence

$$a_i = \int g_i(u) Q_i^u(C_i(F)) du \geq 0.99 \cdot 0.9 c_t =: a_t > 0.$$

At a successful trajectory with midpoint u , the proposal density relative to P_i is

$$\frac{a_i \mathbf{1}_U(u)}{|U| g_i(u) Q_i^u(C_i(F))} \leq \frac{1}{0.98 \cdot 0.9 c_t},$$

since $a_i \leq 1$. Taking G to be the maximum of two and this constant proves Equation (7.9); its final identity holds because both measures are probabilities.

For collision with the fixed opponent, first use the unconditioned bridge B_i . The event contributing under P_i satisfies the regularity and confinement cutoffs. Lemma 6.3 bounds its probability by $C(tr + rD^{3\gamma}h_0^{-1/2})$. Dividing by $a_i \geq a_t$ proves the required estimate for P_i . For the proposal, integrate that lemma's two-leg estimate over the whole target cell; restricting the integral to U can only decrease it. Division by $|U| \geq 0.98$ and by the through- u success probability, which is at least 0.9, proves the proposal estimate. The endpoint separation required in that lemma holds because i was eligible and the opponent has a distinct recorded label.

Finally $\gamma = 1/12$, $h_0 = D^{-1}$ and $r = \alpha/D$ give $rD^{3\gamma}h_0^{-1/2} = \alpha D^{-1/4}$ and $tr = O_t(D^{-1})$. The α -window is fixed. This proves Equation (7.10), with constants depending on the already fixed t but not on the later choice of M or the dilution threshold. $\square$

We record explicitly how Proposition 7.4 applies when two proposals share a midpoint. Suppose that i and j are accessible labels for which B_w is a target cell, and put $U = T_i(w) \cap T_j(w)$. Then $|U| \geq 0.98$. Choose one point uniformly in U and, conditionally on that point, choose two independent successful paths, one with each label's bridge law. Each path separately has law Π_i^U or Π_j^U . Thus each marginal has density at most G relative to its own P -law, and each has collision probability at most e_D against any fixed regular, confined opponent of a distinct label. These bounds integrate the common midpoint. They do not assert a collision bound conditionally on its value: choosing the midpoint at an opponent's midpoint would force a collision. Nor do we assert a product density bound for the pair of paths, whose midpoints agree. If the two paths are used in two separately constrained configurations, there is no avoidance check between them. Union bounds for their separate checks therefore use only the two marginal estimates just proved. A factor consisting of one successful path together with its midpoint has path law Π_i^U and midpoint density at most $1/0.98 \leq G$ relative to uniform measure on B_w .

Each changed path has only CM potential opponents by confinement. Consequently a fixed number of separately constrained configurations has a one-variable failure bound of the form CMe_D . The comparison argument will choose M after G , then increase D so that this quantity is small. In the second-moment calculation, conditional normalizers will cancel away from encounters between the two lattice routes. This is what makes the final likelihood bound uniform in their lengths.

8 Paths through a sparse set of obstacles

The exposed groups supply a random set of usable cells, with a joint exponential bound on prescribed failures. We must connect distant usable cells by random routes whose pairwise encounters remain bounded in exponential mean, uniformly in the volume. This section proves the required statement for an arbitrary dependent random bad set. It combines the dyadic skeleton construction of OpenAI [35, Section 8.1, Lemma 8.1] with the nearest-neighbor detours and component expansion of OpenAI [36, Section 7, Proposition 7.1]. We reproduce both arguments; only the classical boundary-connectivity theorem stated below is used without proof.

Write $\mathcal{L}_K = (\mathbb{Z}/K\mathbb{Z})^3$, with periodic maximum distance d_∞ . Nearest-neighbor edges join vertices differing by one in one coordinate; $*$ -edges join distinct vertices at maximum distance one. A random datum $\mathcal{D}$ determines a bad set $\mathcal{B} \subset \mathcal{L}_K$. All other vertices are called good. Our assumption is

$$\mathbb{P}(A \subset \mathcal{B}) \leq p^{|A|} \quad (A \subset \mathcal{L}_K). \quad (8.1)$$

An ordered pair (v, w) is admissible if the forward displacement of its first coordinate has an integer representative n with $\lceil K/10 \rceil \leq n \leq \lfloor K/8 \rfloor$, and its two transverse displacements have representatives d_2, d_3 satisfying $|d_i| \leq n$. These representatives determine the lift displacement (n, d_2, d_3) uniquely. Set $v = 1/2000$. For two simple paths π, π' define

$$J_R(\pi, \pi') = \sum_{z \in \pi} \sum_{z' \in \pi'} \mathbf{1}_{\{d_\infty(z, z') \leq R\}}. \quad (8.2)$$

Here a simple path visits each vertex at most once.

Proposition 8.1 (Good paths with bounded encounters). *Fix an integer $R \geq 1$. There are constants $p_0, \kappa > 0$, $C_{\text{path}} < \infty$, and an integer K_0 , depending only on R , with the following property. Suppose $K \geq K_0$ and (8.1) holds with $p \leq p_0$. For every admissible ordered pair (v, w) there is a $\mathcal{D}$ -measurable event $\mathcal{E}_{v,w}$ of probability at least $1/2$. On this event there is a measurable probability kernel on simple good nearest-neighbor paths from v to w . Two independent draws from this kernel, conditional on $\mathcal{D}$, satisfy*

$$\mathbb{E} \left[\mathbf{1}_{\mathcal{E}_{v,w}} \mathbb{E} \left[e^{\kappa J_R(\pi, \pi')} \mid \mathcal{D} \right] \right] \leq C_{\text{path}}. \quad (8.3)$$

The constants are uniform in the datum, its law, and the endpoint pair. For every sufficiently large K , at least vK^6 ordered pairs are admissible.

The event is determined before drawing the paths. The expectation in (8.3) averages over one common environment and two fresh path samples; it is not a bound for every fixed environment. We will use the proposition with $R = 30$.

8.1 A folded random skeleton

The skeleton first advances in one coordinate while fluctuating in the other two. Its fluctuations are read forward and then backward, which fixes both endpoints without conditioning on a future value. Independent dyadic blocks make the future transverse displacement sufficiently unpredictable to give summable encounter probabilities. The use of unpredictability and intersection moments originates in Benjamini et al. [6, Theorem 1.3 and Lemma 3.1]; the aligned block construction is related to Häggström and Mossel [25, Section 3, Proposition 3.1]. Finite-endpoint path averaging appears in Abbe et al. [1, Section 6, Lemma 6.1] and Garban and Spencer [24, Section 2, Lemma 2.5]. The following explicit law and the two estimates are those of OpenAI [35, Section 8.1].

Fix an admissible pair, choose a lift of v , and give w the lift $v + (n, d_2, d_3)$. For each $j \in \{2, 3\}$ use independent random variables $U_{\ell, k}^{(j)}$, uniform on $[-1, 1]$, for integers $\ell, k \geq 0$. Put

$$v_i^{(j)} = c_{\text{vel}} \sum_{\ell \geq 0} 2^{-\ell/4} U_{\ell, \lfloor (i-1)/2^\ell \rfloor}^{(j)}, \quad Z_t^{(j)} = \sum_{i=1}^t v_i^{(j)}, \quad Z_0^{(j)} = 0. \quad (8.4)$$

Fix $c_{\text{vel}} = (1 - 2^{-1/4})/2$, so that $c_{\text{vel}} \sum_{\ell \geq 0} 2^{-\ell/4} = 1/2$. The series converges absolutely for every sample. The base vertices are

$$u_t = v + \left(t, \left\lfloor td_2/n + Z_{\min(t, n-t)}^{(2)} \right\rfloor, \left\lfloor td_3/n + Z_{\min(t, n-t)}^{(3)} \right\rfloor \right), \quad 0 \leq t \leq n. \quad (8.5)$$

The endpoints are v, w . Each base increment advances the first coordinate by one and changes each transverse coordinate by at most two. Refine it by first making the first-coordinate step, then all required steps in coordinate two, then those in coordinate three. This takes at most five nearest-neighbor steps. The occurrence list consists of the initial vertex and the vertex reached after each step, without adding an extra copy at the join between consecutive refinements. Denote the projected walk by Γ and retain this lift for the detour construction.

For two independent copies Γ, Γ' of this law and a fixed integer $R_s \geq 1$, define the occurrence count

$$J_{\text{occ}}^{(R_s)} = \sum_{z \in \Gamma} \mathbf{1}_{\{d_\infty(z, \Gamma') \leq R_s\}} + \sum_{z' \in \Gamma'} \mathbf{1}_{\{d_\infty(z', \Gamma) \leq R_s\}}. \quad (8.6)$$

Both sums count occurrences, including any repetitions. For a fixed $B > 0$, put

$$S_B(\Gamma, \Gamma') = \sum_{z \in \Gamma} e^{-d_\infty(z, \Gamma')/B},$$

again counting occurrences.

Lemma 8.2 (Skeleton encounters). *For two independent skeletons just constructed, there are absolute constants $c_{\text{hit}} > 0$ and $C_{\text{hit}} < \infty$ such that, for every integer $R_s \geq 1$,*

$$\mathbb{E} e^{\alpha_s J_{\text{occ}}^{(R_s)}} \leq C_{\text{hit}}, \quad \alpha_s = \frac{c_{\text{hit}}}{(1 + R_s)^2}. \quad (8.7)$$

For every fixed $B > 0$, there are $\xi(B) > 0$ and $C(B) < \infty$ such that

$$\mathbb{E} e^{\xi(B) S_B(\Gamma, \Gamma')} \leq C(B). \quad (8.8)$$

Both estimates are uniform in K and the admissible endpoint pair.

Proof. Write $Z_t = (Z_t^{(2)}, Z_t^{(3)})$ and let Z'_t be the corresponding sum for the second skeleton. Put $h = \lfloor n/2 \rfloor$. Reveal the entire second skeleton at the outset. For the first skeleton, let $\mathcal{F}_t$ contain the second array and precisely those variables $U_{\ell, k}^{(j)}$ whose index blocks

$$I_{\ell, k} = \{k2^\ell + 1, \dots, (k+1)2^\ell\}$$

meet $\{1, \dots, t\}$, for $j = 2, 3$. This filtration includes all variables used through time t , including large blocks continuing beyond t .

For $k \geq 1$ and $t + k \leq h$, the interval $\{t + 1, \dots, t + k\}$ contains a full aligned dyadic block of length $L \geq k/4$. For $k \geq 2$, choose a power of two with $k/4 < L \leq k/2$; aligning its start skips fewer than L indices. For $k = 1$, use a singleton. The variable attached to this block is independent of

$\mathcal{F}_t$ and occurs in $Z_{t+k}^{(j)}$ with coefficient $c_{\text{vel}}L^{3/4}$. Conditional on every other variable, it therefore supplies a uniform density bounded by $Ck^{-3/4}$. Choose this unused variable separately in the two independent transverse coordinates. Integrating the remaining variables proves the conditional two-dimensional density bound

$$\|\text{dens}(Z_{t+k} \mid \mathcal{F}_t)\|_{\infty} \leq Ck^{-3/2}. \quad (8.9)$$

The exponent $3/2 > 1$ is what makes the sum over future times finite.

We next reduce encounters anywhere on the folded walks to tests at the same unfolded time. Let $D_t = |Z_t - Z'_t|_{\infty}$, $0 \leq t \leq h$. An occurrence z at first-coordinate level t is at maximum distance at most two from the base vertex u_t . For an opposing occurrence z' at level t' , write $d = d_{\infty}(z, z')$. Since both first-coordinate spans have length $n \leq K/8$, their differences do not wrap, and $|t - t'| \leq d$. Each base step has maximum norm at most two. Thus, with u'_t the second base vertex,

$$d_{\infty}(u_t, u'_t) \leq 2 + d + 2 + 2|t - t'| \leq 3d + 4.$$

At equal times the linear drifts cancel. Each random deviation has absolute value at most $n/4$, so the lifted transverse difference, including rounding, is less than $K/2$ for our sufficiently large K . It is consequently the ordinary rather than a wrapped difference. Rounding costs at most one, and hence, for $u = \min(t, n - t)$,

$$D_u \leq 3d_{\infty}(z, z') + 5. \quad (8.10)$$

There are at most five occurrences at each first-coordinate level and at most two levels with a given value of $\min(t, n - t)$. Applying (8.10) in both directions gives

$$J_{\text{occ}}^{(R_s)} \leq C \sum_{t=0}^h \mathbf{1}_{\{D_t \leq C(R_s+1)\}}, \quad (8.11)$$

$$S_B(\Gamma, \Gamma') \leq C(B) \sum_{t=0}^h e^{-D_t/(3B)}. \quad (8.12)$$

For the second estimate, take z' closest to z in (8.10) and use $e^{-d/B} \leq e^{5/(3B)} e^{-D_u/(3B)}$.

Let $X_t \in [0, 1]$ be either test on the right of these displays. It is $\mathcal{F}_t$ -measurable. Integrating (8.9) over a square, or against the decaying exponential on $\mathbb{R}^2$, and summing $k^{-3/2}$ gives

$$\mathbb{E} \left[\sum_{u=t+1}^h X_u \mid \mathcal{F}_t \right] \leq b_0, \quad b_0 = \begin{cases} C(R_s + 1)^2, & \text{for the indicator test,} \\ C(B), & \text{for the exponential test.} \end{cases} \quad (8.13)$$

Set $B_0 = b_0 + 1$. For every $j \geq 1$, successive conditioning of the last index in the ordered sum proves

$$\mathbb{E} \sum_{0 \leq t_1 < \dots < t_j \leq h} X_{t_1} \cdots X_{t_j} \leq B_0^j.$$

For $j = 1$, this follows from (8.13) at time zero and $X_0 \leq 1$; each subsequent conditioning costs at most b_0 . Since $e^{\theta x} - 1 \leq (e^{\theta} - 1)x$ for $0 \leq x \leq 1$,

$$\mathbb{E} e^{\theta \sum_{t=0}^h X_t} = \mathbb{E} \prod_{t=0}^h (1 + e^{\theta X_t} - 1) \leq \sum_{j \geq 0} ((e^{\theta} - 1)B_0)^j.$$

Taking θ a sufficiently small constant times B_0^{-1} bounds this geometric series by two. Equations (8.11)–(8.12) give the two asserted moments, after adjusting their constants. $\square$

The skeleton law and its bounds use no many-particle parameter. In applying them, we draw the two arrays independently of each other and of the exposed datum $\mathcal{D}$.

For later use we convert the occurrence count to our pair count. Every first-coordinate level of a lifted skeleton contains at most five occurrences: level zero has only the initial vertex, and all vertices created in the t th refinement have first coordinate $v_1 + t$. The first-coordinate span is $n \leq K/8$, so the periodic distance between two such levels is their ordinary distance. A fixed occurrence of Γ can therefore have at most $5(2R + 1)$ opposing occurrences within distance R . With $R_s = R$, this proves

$$J_R(\Gamma, \Gamma') \leq 5(2R + 1)J_{\text{occ}}^{(R)}. \quad (8.14)$$

The left side uses the double occurrence sum from Equation (8.2). This conversion uses the actual refinement law, as well as its moment bounds.

8.2 Finite components and their exterior boundaries

We record the deterministic facts used to move a lattice path around bad vertices. For a finite $*$ -connected nonempty set $C \subset \mathbb{Z}^3$, let $\text{fill}(C)$ consist of C and all finite nearest-neighbor components of $\mathbb{Z}^3 \setminus C$. The remaining component is the exterior. There is only one infinite component: outside any box containing C , nearest-neighbor vertices communicate, and every infinite component meets that region.

Lemma 8.3 (Exterior shell). *The set $\text{fill}(C)$ lies in the coordinate bounding box of C . The exterior vertices at maximum distance one from C form a nearest-neighbor connected set, denoted $\partial_*^{\text{ext}} C$, with at most $26|C|$ vertices. If C is a $*$ -component of a larger bad set, every vertex of this shell is good.*

Proof. A vertex outside the coordinate box has a coordinate ray to infinity that misses C , so it is not in a finite complementary component. For the shell's connectivity we apply Kesten's boundary-connectivity theorem in the formulation of Timár [40, arXiv version 2, Theorem 4]: for a finite $*$ -connected subset of $\mathbb{Z}^3$, its exterior $*$ -neighbors, accessible from infinity through its nearest-neighbor complement, are nearest-neighbor connected. These are exactly the vertices in the shell just defined. There are at most 26 possible $*$ -neighbors per vertex of C . A bad shell vertex would be $*$ -adjacent to C and hence belong to the same bad component, which is impossible for a vertex outside C . $\square$

For an integer $a \geq 1$, range- a adjacency joins distinct torus vertices whose maximum distance is at most a .

Lemma 8.4 (Small periodic components). *Suppose $K > 2a$, and a range- a component on $\mathcal{L}_K$ has m vertices with $am < K$. Each of its lifts to $\mathbb{Z}^3$ is a finite component projecting bijectively onto it. The coordinate span of a lift is at most $a(m - 1)$.*

Proof. Choose a spanning tree, lift its root, and lift each tree edge by its unique displacement in $[-a, a]^3$. Distinct torus vertices have distinct lifts. The difference between any two lifted vertices has maximum norm at most $a(m - 1)$, by their tree path. For a remaining edge, its assigned endpoint difference minus its short displacement belongs to $K\mathbb{Z}^3$ and has maximum norm at most $am < K$. It must therefore vanish. All edges are consistent with these lifts. The full inverse image is their disjoint translates by $K\mathbb{Z}^3$; no edge connects two translates. $\square$

We will also use a basic counting bound. For each fixed range a , there is a constant A_a such that the number of connected sets of m vertices containing a specified root is at most A_a^m , on either $\mathbb{Z}^3$

or the torus. To see this, choose deterministically a rooted spanning tree for each set and traverse it depth first. The traversal has $2(m-1)$ steps, each chosen from a fixed finite set of possible displacements, and its visited vertices recover the set. Increasing A_a covers $m=1$. Together with (8.1), this gives

$$\mathbb{P}(\text{a range-}a \text{ bad component has at least } m \text{ vertices}) \leq K^3(A_ap)^m. \quad (8.15)$$

A component of size at least m contains a connected subset of exactly m vertices, by growing a spanning tree one vertex at a time.

We have proved the two encounter bounds for paths that ignore the bad set. To use them, we will retain the skeleton vertices whenever possible and replace obstructed segments by short exterior routes. A size restriction guarantees the finite lifts needed for this construction; the endpoints must also lie outside the components' filled interiors. The joint bad-set bound makes both requirements likely.

Fix $a_1 = R + 2$. Reject the environment if it contains a range- a_1 bad component of size at least

$$m_K = \left\lfloor \frac{K}{10a_1} \right\rfloor.$$

For large K , every remaining bad $*$ -component has finite periodic lifts, by Lemma 8.4. Reject also if either endpoint belongs to the projection of the fill of one of these components. The event on which neither rejection occurs is $\mathcal{E}_{v,w}$.

If the fill of a lifted $*$ -component of size m contains a lift of an endpoint, some root of the component lies within maximum distance m of that endpoint. This follows from the bounding-box assertion and the coordinate span in Lemma 8.4. There are at most Cm^3 possible roots modulo K . Counting connected sets and using (8.1) thus yields

$$\mathbb{P}(\mathcal{E}_{v,w}^c) \leq K^3(A_{a_1}p)^{m_K} + C \sum_{m \geq 1} m^3(A_1p)^m. \quad (8.16)$$

The constant includes both endpoints. First make p_0 small enough that the series is at most $1/4$, then make K_0 large enough that the first term is at most $1/4$ for $p \leq p_0$ and $K \geq K_0$. This proves $\mathbb{P}(\mathcal{E}_{v,w}) \geq 1/2$. We may decrease p_0 further below.

8.3 Nearest-neighbor detours

Draw the skeleton of Section 8.1 independently of the datum. Its folding in Equation (8.5) fixes the final endpoint without conditioning the random process on a future value. We now adapt this nearest-neighbor walk to the bad set while keeping the endpoint rejection event fixed.

Figure 1 illustrates the replacement. The construction takes place in $\mathbb{Z}^3$, where a finite obstacle has a well-defined exterior; projection and loop erasure are performed afterward.

On $\mathcal{E}_{v,w}$, replace the visits of this lifted walk to bad components as follows. If it meets a lifted bad $*$ -component C , take its first and last visits to C . The vertices immediately before and after these visits belong to $\partial_*^{\text{ext}}C$: the path from the first endpoint to the first such vertex misses C , and the path from the last such vertex to the other endpoint misses C ; both endpoints are outside $\text{fill}(C)$. By Lemma 8.3, join these two shell vertices by a simple nearest-neighbor path in the shell, replacing the intervening part of the walk. The added vertices are all good. Each replacement removes at least one bad visit, and no replacement introduces a bad visit. The process therefore terminates after finitely many steps.

Every component treated in this procedure was met by the original nominal refinement: any remaining bad vertex still belongs to that original walk. Project the final walk to the torus and

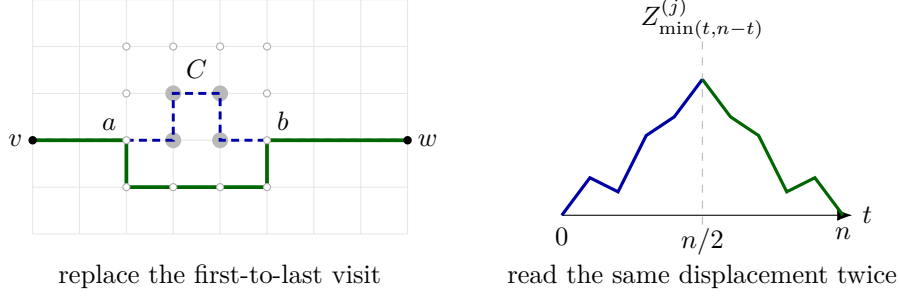


Figure 1: Two parts of the path construction, shown schematically. Left: the original dashed route meets the finite bad component C ; the solid route joins its exterior predecessor a to its exterior successor b through good nearest-neighbor vertices. The drawing is a planar schematic of a three-dimensional exterior shell, and does not represent holes or later loop erasure. Right: for even n , the random transverse displacement used in the skeleton is reflected at $n/2$; the common linear drift and integer rounding are omitted. This return to zero fixes the final endpoint without a bridge conditioning.

erase loops in chronological order. The result is a simple good nearest-neighbor path from v to w . Every nonnominal vertex belongs to the shell of a bad $*$ -component met by Γ . All choices in the refinement, shell paths, and erasure can be made using fixed orderings. Thus the construction gives the measurable kernel in Proposition 8.1.

We now bound the encounters introduced by this construction. Use two independent nominal samples Γ, Γ' in the same environment, and let π, π' be their final simple paths. Let H range over the range- a_1 bad components on the torus. Fix $C_R = 52(2R + 1)^3$, which also exceeds $5(2R + 1)$ in Equation (8.14). On $\mathcal{E}_{v,w}$,

$$J_R(\pi, \pi') \leq J_R(\Gamma, \Gamma') + C_R \sum_H |H| \mathbf{1}_{\{d_\infty(H, \Gamma) \leq a_1, d_\infty(H, \Gamma') \leq a_1\}}. \quad (8.17)$$

For nominal walks, J_R here counts vertex occurrences, which only increases the bound.

To prove (8.17), first count encounters where both vertices are nominal. These are bounded by the first term. If a nonnominal vertex belongs to the shell of C and encounters a nominal vertex of the other walk, then C is within distance $R + 1 \leq a_1$ of that other walk, and C meets its own nominal walk. If two nonnominal vertices belong to shells of C, C' , then $d_\infty(C, C') \leq R + 2 \leq a_1$, so both components belong to one range- a_1 component H meeting both nominal walks. Charge each such encounter to one of its nonnominal vertices and to this H . There are at most $26|H|$ possible added vertices associated to the $*$ -components inside H , even if several lifted copies were used: projection can only decrease this number. Each such vertex encounters at most $(2R + 1)^3$ vertices of the other simple path. Charging from either path costs at most an additional factor two. This proves the claimed bound.

8.4 Averaging the environment and applying both moments

Fix both complete nominal paths. Their independence from $\mathcal{D}$ leaves (8.1) available for any prescribed collection of bad vertices. Expand the exponential of the second term in (8.17) as a product over the actual range- a_1 components. A term corresponding to distinct components has

disjoint vertex sets. Drop maximality, sum over all disjoint connected candidate sets, and apply (8.1). Finally drop the disjointness restriction and use $1 + x \leq e^x$. The resulting bound is

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} \left[\mathbf{1}_{\mathcal{E}_{v,w}} \exp \left(\kappa C_R \sum_H |H| \mathbf{1}_{\{d_\infty(H, \Gamma), d_\infty(H, \Gamma') \leq a_1\}} \right) \right] \\ \leq \exp \left(\sum_F (p e^{\kappa C_R})^{|F|} \mathbf{1}_{\{d_\infty(F, \Gamma), d_\infty(F, \Gamma') \leq a_1\}} \right), \end{aligned} \quad (8.18)$$

where F ranges over all nonempty range- a_1 connected torus sets. The factor $e^{\kappa C_R |F|} - 1$ from the expansion was bounded above by $e^{\kappa C_R |F|}$. Thus only joint bad-set probabilities were used; there is no independence assumption on the environment.

We bound the exponent by the soft-distance statistic in Equation (8.8). If a contributing set F has size m , choose an occurrence $z \in \Gamma$ within distance a_1 of a root of F . There are at most $(2a_1 + 1)^3$ root choices for each z , and at most $A_{a_1}^m$ connected sets at each root. A spanning-tree path in F has at most $m - 1$ edges, each of length at most a_1 in the periodic maximum metric. Since F is also within a_1 of Γ' , the triangle inequality gives

$$d_\infty(z, \Gamma') \leq 2a_1 + a_1(m - 1) = a_1(m + 1).$$

This argument applies even to wrapping candidate sets; only the actual components used for detours required the earlier lift and rejection test. Absorb the root choices into a constant $A_R \geq 1$. The exponent in Equation (8.18) is at most

$$\sum_{z \in \Gamma} \sum_{m \geq 1} q^m \mathbf{1}_{\{d_\infty(z, \Gamma') \leq a_1(m+1)\}}, \quad q = A_R p e^{\kappa C_R}. \quad (8.19)$$

All constants here depend only on the fixed R . Counting the same set more than once only increases this nonnegative upper bound.

Require $q \leq e^{-2}$. If $d \leq a_1(m + 1)$, split q^m into two equal powers and use

$$q^m = q^{m/2} q^{m/2} \leq q^{m/2} e^{-m} \leq e q^{m/2} e^{-d/a_1}.$$

Summing the first factor over $m \geq 1$ proves

$$\sum_{m \geq 1} q^m \mathbf{1}_{\{d \leq a_1(m+1)\}} \leq \eta(p, \kappa) e^{-d/a_1}, \quad \eta(p, \kappa) = \frac{e\sqrt{q}}{1 - \sqrt{q}}. \quad (8.20)$$

In particular $\eta(p, \kappa) \rightarrow 0$ as $p \rightarrow 0$ at fixed κ . Set $B = a_1$. Equations (8.17), (8.14), and (8.18)–(8.20) give

$$\mathbb{E}_{\mathcal{D}} \left[\mathbf{1}_{\mathcal{E}_{v,w}} e^{\kappa J_R(\pi, \pi')} \mid \Gamma, \Gamma' \right] \leq \exp \left(\kappa C_R J_{\text{occ}}^{(R)} + \eta(p, \kappa) S_B(\Gamma, \Gamma') \right). \quad (8.21)$$

On the left, the adapted paths are used only on $\mathcal{E}_{v,w}$. The environment was averaged once, with both entire skeletons fixed.

First choose $\kappa > 0$ with $2\kappa C_R \leq \alpha_R$, where $\alpha_R = c_{\text{hit}}/(1 + R)^2$. Next choose $p_0 > 0$ so small that $q \leq e^{-2}$ and $2\eta(p, \kappa) \leq \xi(B)$ for $p \leq p_0$, as well as the earlier endpoint rejection requirements. The constants K_0 can then be enlarged to cover those rejection estimates. Cauchy–Schwarz and both moments in Lemma 8.2 imply

$$\begin{aligned} \mathbb{E} \exp \left(\kappa C_R J_{\text{occ}}^{(R)} + \eta S_B \right) &\leq \left(\mathbb{E} e^{2\kappa C_R J_{\text{occ}}^{(R)}} \right)^{1/2} \left(\mathbb{E} e^{2\eta S_B} \right)^{1/2} \\ &\leq \sqrt{C_{\text{hit}} C(B)} =: C_{\text{path}} < \infty. \end{aligned}$$

Together with Equation (8.21), this proves the annealed bound (8.3). Both path draws use one common datum and independent skeleton seeds. The event retains its original mass; there is no renormalization by conditioning on acceptance and no assertion of a bound for each fixed environment.

Finally, for $K \geq 80$ the interval $[\lceil K/10 \rceil, \lfloor K/8 \rfloor]$ contains at least $K/40 - 1 \geq K/80$ integers. For each such n , there are $(2n+1)^2 \geq K^2/25$ transverse choices. They are distinct modulo K , since $2n < K$. Multiplying by the K^3 choices of starting vertex gives at least $K^6/2000 = \nu K^6$ admissible ordered pairs. This completes Proposition 8.1.

9 A common measure and positive occupation

We now turn the supply of movable labels into a lower bound on the constant-orbital fraction B_{occ} . Recall from Proposition 5.1 that the midpoint configuration X and the retained configuration W' have density $\Phi(X, W')^2$, where Φ is the nonnegative kernel of the square root of the canonical state. The input is the conditional path law of Section 7 and the good lattice routes of Proposition 8.1. Along one route we construct two copies of the selected trajectories. Their midpoint assignments leave the same observed bath after different tags are removed. We then bound the second moment of the resulting change of measure by encounters between two routes.

This is the comparison mechanism of OpenAI [36, Section 8]; its conditional normalizer argument is reproduced below. The thermal modification is to retain W' in every observation and in the exposed data. It is never resampled. All constants obtained here are uniform in D, K and $0 < T \leq D^{-1}$ once the earlier fixed parameters have been chosen.

9.1 Affinity and a common finite measure

For finite nonnegative measures P, Q on the same measurable space, define their affinity by

$$\text{Aff}(P, Q) = \int \sqrt{pq} \, d\xi, \quad p = \frac{dP}{d\xi}, \quad q = \frac{dQ}{d\xi}, \quad \xi = P + Q.$$

Using another dominating measure gives the same value. We will use three elementary properties. If $P \geq P'$ and $Q \geq Q'$, then $\text{Aff}(P, Q) \geq \text{Aff}(P', Q')$. For finite lists of measure pairs,

$$\text{Aff} \left(\sum_j P_j, \sum_j Q_j \right) \geq \sum_j \text{Aff}(P_j, Q_j). \quad (9.1)$$

Indeed, pointwise Cauchy–Schwarz gives $\sum_j \sqrt{p_j q_j} \leq \sqrt{(\sum_j p_j)(\sum_j q_j)}$. Finally, applying the same probability kernel to both measures cannot decrease affinity. To see this, form the joint measures with that kernel and then condition a common dominating measure on the output. Conditional Cauchy–Schwarz gives $\mathbb{E}(\sqrt{pq} \mid \text{output}) \leq \sqrt{\mathbb{E}(p \mid \text{output})\mathbb{E}(q \mid \text{output})}$. Integration proves the assertion. Projection is a special case.

Lemma 9.1 (A common finite measure). *Let P, Q be finite nonnegative measures and let ν be a finite measure of mass $m > 0$, absolutely continuous with respect to both. If*

$$\int \left(\frac{d\nu}{dP} \right)^2 dP \leq C_P, \quad \int \left(\frac{d\nu}{dQ} \right)^2 dQ \leq C_Q,$$

then $\text{Aff}(P, Q) \geq m^2 / \sqrt{C_P C_Q}$.

Proof. With densities p, q, s relative to a common dominating measure, Hölder's inequality with exponents 4, 4, 2 gives

$$m = \int \left(\frac{s^2}{p} \right)^{1/4} \left(\frac{s^2}{q} \right)^{1/4} (pq)^{1/4} \leq C_P^{1/4} C_Q^{1/4} \text{Aff}(P, Q)^{1/2}.$$

The ratios can be set to zero where their denominators vanish, since s is zero there. Squaring and rearranging proves the claim. $\square$

9.2 Cell measures and disjoint groups of tags

Recall $V = K^3$, $D = N/V$, and the unit cells B_v , each of volume one. Write $X' = (x_2, \dots, x_N)$ for the ordered bath. For ordered cells v, w , define finite measures on $(x, y, X', W') \in B_v \times B_w \times \Lambda_K^{N-1} \times \Lambda_K^N$ by

$$\begin{aligned} dP_{vw} &= V \Phi(x, X', W')^2 dx dy dX' dW', \\ dQ_{vw} &= V \Phi(y, X', W')^2 dx dy dX' dW'. \end{aligned} \tag{9.2}$$

Nonnegativity of Φ and the occupation formula in Proposition 5.1 give the exact identity

$$B_{\text{occ}} = \frac{1}{V^2} \sum_{v,w} \text{Aff}(P_{vw}, Q_{vw}). \tag{9.3}$$

Fix a group index $1 \leq g \leq G_D := \lfloor \ell D / (4M) \rfloor$ from Section 7. Its exposed data are $\mathcal{B}$, its unexposed label set is $\mathcal{I}$, and its conditional path law is $\lambda_{\mathcal{I}}$ from Lemma 7.2. Define measures $\hat{P}_g, \hat{Q}_g$ on $(\mathcal{B}, x, y, X')$; the data $\mathcal{B}$ already include W' . For $\hat{P}_g$, sample the data and the conditional unexposed paths, and *sum* the following contributions over all $i \in \mathcal{I}$: take $x = \omega_i(t)$, restrict to $x \in B_v$, order the remaining $N - 1$ midpoints by a fresh uniform permutation to form X' , and sample y uniformly in B_w . For $\hat{Q}_g$, take the tagged midpoint as $y \in B_w$ and sample the extra point x uniformly in B_v . The use of a sum, rather than a uniformly chosen tag, fixes their normalization.

Lemma 9.2 (Tag normalization). *For each ordered pair of cells,*

$$\text{Aff}(P_{vw}, Q_{vw}) \geq \frac{1}{D} \sum_{g=1}^{G_D} \text{Aff}(\hat{P}_g, \hat{Q}_g). \tag{9.4}$$

Proof. Put all groups on the original slab sample, with all allocation randomizations and a fresh uniform permutation of the full label set. After deleting a tag, the restricted permutation uniformly orders its bath. Each group's conditional construction is a disintegration of this common sample with respect to its own exposed data. Thus the differing data spaces are projected away before the following comparison; no conditional symmetry of the unexposed-path law is asserted. For every sampled path configuration, group label sets are disjoint. Their sum of tag contributions is bounded by the sum over all N trajectory labels. The unrestricted midpoint vector together with W' has density $\Phi(X, W')^2$, symmetric in X for each W' . For each fixed tag label, uniform ordering of the bath therefore gives the same density $\Phi(x, X', W')^2$. Summing all tags gives N times that density. Since the extra target cell has volume one and $V/N = 1/D$, projecting to (x, y, X', W') yields

$$\frac{1}{D} \sum_g \text{proj } \hat{P}_g \leq P_{vw}, \quad \frac{1}{D} \sum_g \text{proj } \hat{Q}_g \leq Q_{vw}.$$

Use monotonicity, (9.1), and the increase of affinity under projection. Its homogeneity supplies the factor $1/D$. $\square$

The remaining task is to find, for each retained group and each of many ordered cell pairs, a common measure of mass at least $1/2$ whose squared likelihood integrals against $\hat{P}_g$ and $\hat{Q}_g$ are bounded by one constant C_2 . Lemma 9.1 would then give affinity at least $1/(4C_2)$ per group. The factor $1/D$ in Equation (9.4) can be paid because the number of disjoint groups is proportional to D .

9.3 Two midpoint assignments with the same observed bath

Take $R = 30$ in Proposition 8.1, and fix an admissible ordered cell pair v, w : its lift displacement (n, d_2, d_3) satisfies $\lceil K/10 \rceil \leq n \leq \lfloor K/8 \rfloor$ and $|d_i| \leq n$. Choose the good-site parameters of Proposition 7.3 to meet the bad-set hypothesis of Proposition 8.1. Let $\mathcal{E}_{vw}$ be the data-measurable event supplied by Proposition 8.1, of probability at least $1/2$. On this event draw a simple nearest-neighbor path

$$z_0 = v, z_1, \dots, z_d = w$$

through good sites. For each $k < d$, independently choose a uniform accessible label i_k anchored at z_k . The choices inspect only $\mathcal{B}$ and the lattice path. In particular, they do not inspect any latent trajectory. The selected labels are distinct because their anchors are distinct.

We construct two path configurations that share their observed (x, y, X', W') . Keep the exposed data fixed. First sample all unselected unexposed trajectories from their exact marginal under $\lambda_{\mathcal{I}}$, and use them in both copies. For every vertex z_k , prepare a factor as follows. The first copy needs label i_k through this cell if $k < d$, and the second copy needs label i_{k-1} through it if $k > 0$. Intersect the corresponding good-target sets $T_{i_k}(z_k)$ and $T_{i_{k-1}}(z_k)$ from Section 7, using only the one required set at an endpoint. The intersection, denoted by A_k , has volume at least 0.98. Sample U_k uniformly in A_k , and then independently sample the one or two required bridge trials through U_k , each conditioned on its individual success against the frozen paths. All $d + 1$ vertex factors are independent before the additional conditioning below. Figure 2 shows the two midpoint assignments.

Use U_d as the extra point in the first copy and U_0 as the extra point in the second copy. Impose all remaining hard-avoidance checks in both copies: selected paths must avoid the unselected unexposed paths and one another within their respective copies. Condition the product of vertex factors on these checks, separately for each fixed collection of unselected paths. The sampled unselected-path marginal therefore remains its original marginal under $\lambda_{\mathcal{I}}$. More explicitly, let λ_{out} be that marginal, let F_k be the probability law of the k th vertex factor, and let J be the indicator of the remaining checks in both copies. The joint law just defined is

$$\lambda_{\text{out}}(\mathrm{d}\omega_{\text{out}}) \frac{J(\omega_{\text{out}}, \xi_0, \dots, \xi_d) \prod_{k=0}^d F_k(\mathrm{d}\xi_k)}{\int J(\omega_{\text{out}}, \xi_0, \dots, \xi_d) \prod_{k=0}^d F_k(\mathrm{d}\xi_k)}. \quad (9.5)$$

The denominator is a function of the fixed outside paths. Integrating the second factor gives one for each such configuration. A single normalization after integrating the outside paths would change their marginal and would not yield the comparison used below. Each F_k depends only on the data, route and selected labels, so it is fixed before those outside paths are sampled.

We verify that this conditional construction is defined. A vertex factor carries at most two paths, each with a marginal of the form $\Pi_i^{A_k}$ in Proposition 7.4. Its paths are confined within distance ten of their anchors, and there are at most M unexposed labels per anchor. Consequently each factor has at most $C_{\text{col}}M$ possible opponents in the two copies, for a fixed geometric constant C_{col} . All opponents have distinct labels within the copy in which a check is imposed. The deterministic-opponent

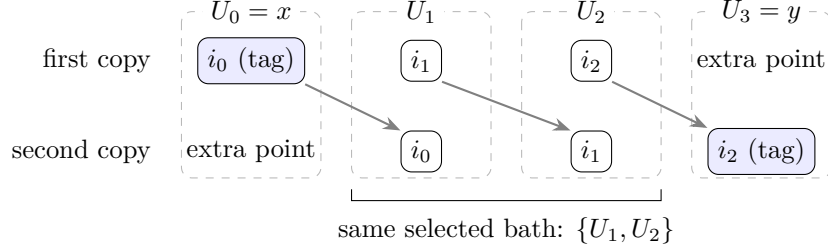


Figure 2: The midpoint assignment for a route of three edges. A dashed column is one vertex factor with one common midpoint. Arrows identify the same particle label in the two copies; they are not trajectories in time. Removing the highlighted tags leaves the same two selected bath midpoints. All unselected bath paths and the configuration W' are shared. For a route of arbitrary length the interior columns continue in the same way.

estimate, summed over those checks, gives

$$\chi_D = C_{\text{col}} M e_D \longrightarrow 0 \quad (D \longrightarrow \infty, M \text{ fixed}). \quad (9.6)$$

For any fixed regular and confined values of the other paths with their recorded endpoints, the factor passes every incident check with probability at least $1 - \chi_D$. This statement averages its common midpoint over A_k ; it makes no claim at a prescribed midpoint. It uses the two path marginals and a union bound, so no independence of the paths within the factor is required. Inserting the $d + 1$ factors successively now bounds the two-copy conditioning probability below by $(1 - \chi_D)^{d+1} > 0$ for sufficiently large D .

Tag i_0 in the first copy and i_{d-1} in the second. In each copy the tag and the extra point then give $x = U_0, y = U_d$. The selected bath midpoints in both copies are exactly $U_1, \dots, U_{d-1}$; the unselected midpoints also agree. Choose a common uniform ordering of the shared midpoint instances; their physical labels in the two copies may differ. In each copy separately, this has the uniform ordering law determined by its paths and its tag. This defines a single finite measure $\hat{\nu}$ on $(\mathcal{B}, x, y, X')$ after averaging the data, routes and labels, with zero mass off $\mathcal{E}_{vw}$. Each conditional sampling law on this event is a probability measure, so

$$\hat{\nu}(\text{all}) = \mathbb{P}(\mathcal{E}_{vw}) \geq \frac{1}{2}. \quad (9.7)$$

These are measurable probability kernels: the route and label choices are finite, the target sets are measurable functions of the data, and all path restrictions are Borel subsets of the continuous-path spaces. The positive normalizers just proved permit the displayed conditioning.

9.4 Cancellation of conditional normalizers

The midpoint construction has two path copies in each internal vertex factor. To estimate its likelihood, designate one copy as primary and regard the other as auxiliary. At an internal vertex z_k , $1 \leq k < d$, the primary path has label i_k in the first-copy view, whereas the auxiliary path has label i_{k-1} . Thus deleting a factor is not the same operation as deleting all instances of one particle label. The following lemma keeps this distinction explicit and isolates the normalizers that cancel when two routes are far apart. Its coordinates and factors are abstract; we identify them with the midpoint construction immediately after the proof. The two changes arise because the square of a route-averaged likelihood expands into a product for two independent route choices at the same exposed datum.

The next lemma, from OpenAI [36, Lemma 8.3], concerns finitely many coordinates in standard Borel spaces, each with a probability reference P_i . All auxiliary spaces below are standard Borel as well. A *check* will mean an indicator imposing a constraint on one or two coordinates. The joint reference law λ is the product $\bigotimes_i P_i$ conditioned on a finite collection of checks. An additional independent coordinate is allowed; it simply has no checks.

A change indexed by a set S leaves the marginal outside S unchanged. Inside S , it starts from independent factors $\Pi_{S,i}$, $i \in S$. Each factor includes a *primary coordinate* in the space of P_i and possibly auxiliary coordinates. Its primary marginal is $g_{S,i}P_i$, with

$$0 \leq g_{S,i} \leq G, \quad \int g_{S,i} dP_i = 1. \quad (9.8)$$

The factors are fixed before the outside coordinates are sampled. The product is conditioned on specified checks, including every reference check involving the primary coordinates in S . Checks involving only fixed outside coordinates are already satisfied almost surely under the unchanged outside marginal; otherwise the conditional proposal would not be defined there. Omit these redundant checks. Every remaining proposal check therefore involves a coordinate carried by a changed factor. Auxiliary coordinates are then forgotten. The resulting primary likelihood relative to λ is denoted by L_S .

For each primary and auxiliary coordinate, fix a measurable allowed set on which all its relevant reference and proposal marginals are concentrated. These sets, like the proposal factors, are fixed before sampling outside coordinates. A configuration is *individually permitted* when each coordinate lies in its allowed set; this imposes no pairwise checks. The compatibility hypothesis below is uniform on these sets, including configurations that fail some pairwise checks.

We make explicit the deletion convention for this construction. When some factors are removed, remove their coordinates and all checks incident to them. Do not replace a removed auxiliary coordinate by an original reference coordinate. Coordinates fixed outside the sets under discussion remain as obstacles in all the checks in which they occur. Coordinates here are path instances in a specified copy. If an auxiliary with label j belongs to a retained factor, it remains integrated in that factor even when a different factor carrying the primary label j is deleted. It is never replaced by the original path with label j .

Lemma 9.3 (Cancellation away from an overlap). *Consider two changes S, T of the preceding kind, with $G \geq 1$, and put $U = S \cup T$. Suppose that the following conditions hold.*

- (i) *There is $\chi \in [0, 1)$ with two insertion bounds. In the reference system, inserting any coordinate $i \in U$ from P_i satisfies all reference checks incident to i with probability at least $1 - \chi$. In the proposal system of either change, inserting a whole factor from $\Pi_{S,i}$ or $\Pi_{T,i}$ satisfies all checks of that proposal incident to its carried coordinates with probability at least $1 - \chi$. Each bound is uniform in every fixed choice of the other individually permitted coordinates of the respective system, and remains valid after deleting checks or coordinates as specified above.*
- (ii) *There are decompositions $S = C_S \sqcup F_S$ and $T = C_T \sqcup F_T$, with $F_S \cap T = F_T \cap S = \emptyset$. No coordinate carried by an F_S factor has a check with any coordinate carried by a T factor, or with the original primary coordinates indexed by T ; the corresponding assertion holds with S, T interchanged. The reference coordinates in F_S, F_T also have no mutual checks.*

Then

$$\int L_S L_T d\lambda \leq G^{|S \cap T|} (1 - \chi)^{-2(|C_S| + |C_T|)}. \quad (9.9)$$

It is enough that the assumptions hold for almost every fixed admissible configuration outside U .

Proof. Put $a_\chi = 1 - \chi$ and $c = |C_S| + |C_T|$. Fix the primary coordinates outside U . All normalizers below are conditional on these fixed values; we first work on a configuration for which this conditional law is defined.

For any selected set, its reference normalizer is the integral of its reference-check indicator against its product references. In particular, write Z_U for this integral over U . When the coordinates outside S are fixed, write Z_S for the corresponding integral over S . Thus Z_S can depend on coordinates in $T \setminus S$. Write Y_S for the integral of the proposal-check indicator J_S against $\bigotimes_{i \in S} \Pi_{S,i}$, with the same outside values fixed.

Disintegrate each factor in the form

$$\Pi_{S,i}(dx_i d\zeta_i) = P_i(dx_i)K_{S,i}(x_i, d\zeta_i), \quad K_{S,i}(x_i, \text{all}) = g_{S,i}(x_i).$$

Here ζ_i denotes all its auxiliary coordinates. Define

$$H_S(x_S; x_{S^c}) = \int J_S \prod_{i \in S} K_{S,i}(x_i, d\zeta_i).$$

On the reference-admissible configurations, the exact likelihood is

$$L_S = \frac{Z_S}{Y_S} H_S. \quad (9.10)$$

Indeed, the conditional reference density is its check indicator divided by Z_S , whereas the new primary density is H_S/Y_S , both relative to $\bigotimes_{i \in S} P_i$. The outside marginal is identical. The inclusion of primary reference checks ensures that the new density is zero on reference-forbidden configurations.

Now retain only the factors indexed by F_S , together with coordinates fixed outside U , using the stated deletion convention. Denote their reference normalizer, proposal normalizer and primary proposal density by $Z_{F_S}, Y_{F_S}, H_{F_S}$. Define the three corresponding F_T quantities using the factors from the T change. These quantities depend only on the outside of U and, for H_{F_S}, H_{F_T} , on their own primary coordinates. They do not depend on the omitted near coordinates.

Successive insertion using assumption (i), and noninteraction in assumption (ii), give

$$Z_U \geq a_\chi^c Z_{F_S} Z_{F_T}, \quad (9.11)$$

$$Z_S \leq Z_{F_S}, \quad Y_S \geq a_\chi^{|C_S|} Y_{F_S}, \quad (9.12)$$

$$Z_T \leq Z_{F_T}, \quad Y_T \geq a_\chi^{|C_T|} Y_{F_T}. \quad (9.13)$$

For the first inequality, the reference integral over the two far sets factorizes. The remaining distinct coordinates form $U \setminus (F_S \cup F_T)$, a subset of $C_S \cup C_T$ with cardinality at most c . Insert them one at a time with the uniform compatibility bound. For the upper bound on Z_S , delete the checks involving C_S and integrate those reference coordinates freely. Coordinates in $T \setminus S$ have no checks with the far set. For its proposal lower bound, start with the reduced F_S integral and add the C_S factors. The fixed original coordinates in $T \setminus S$ still do not meet the far factors. An auxiliary instance carried by a retained factor remains integrated there, even if a newly inserted factor carries its primary label; only the checks of the S proposal are restored. Each addition retains at least the factor a_χ . Applying these two arguments to the T change proves the last line. These statements are pointwise in the opposing coordinates; they do not assert that Z_S or Y_S is constant in them.

The insertion bounds also prove positivity of the divisors. In particular, integrating coordinates one at a time gives a positive product lower bound for every reduced normalizer. The full reference conditional is defined on the outside configurations under consideration.

Deleting proposal checks involving a near factor gives

$$H_S \leq H_{F_S} \prod_{i \in C_S} g_{S,i}, \quad H_T \leq H_{F_T} \prod_{i \in C_T} g_{T,i}.$$

After dropping the reference admissibility indicator in an upper bound, the two far primary integrals equal Y_{F_S} and Y_{F_T} . The remaining coordinates carry only the displayed marginal densities. A coordinate occurring once integrates to one. A coordinate occurring in both lists costs at most G , by (9.8). Therefore

$$\int H_S H_T \mathbf{1}_{\text{admissible}} \prod_{i \in U} P_i(\mathrm{d}x_i) \leq Y_{F_S} Y_{F_T} G^{|S \cap T|}. \quad (9.14)$$

The conditional reference law on U has normalizer Z_U . Combine (9.10)–(9.14). The two pointwise ratio bounds contribute $a_\chi^{-c} Z_{F_S} Z_{F_T} / (Y_{F_S} Y_{F_T})$; division by Z_U contributes at most $a_\chi^{-c} / (Z_{F_S} Z_{F_T})$. All four far normalizers cancel, leaving $G^{|S \cap T|} a_\chi^{-2c}$. Finally integrate the outside coordinates. This proves (9.9). $\square$

9.5 The likelihood cost is confined to encounters

Fix the data, the route and the selected labels. Regard either copy as the primary one. The reference for that copy is $\lambda_{\mathcal{I}} \otimes P_*$, where P_* is the uniform law of the extra point in its endpoint cell. Let

$$S = \{i_0, \dots, i_{d-1}\} \cup \{*\}.$$

In Lemma 9.3, a primary coordinate is the chosen copy's path or endpoint extra point. Its auxiliary coordinates are the other copy's path or extra point through the same midpoint. The checks are the collision exclusions within each copy; there is no collision check between paths in different copies. Index a vertex factor by its primary label, using $*$ for the extra point; its center is its midpoint cell z_k . The two paths in one factor may have different labels, and each copy contributes at most one path. The allowed set for each path consists of the regular, confined, frozen-avoiding paths with its own recorded endpoints. Each extra point lies in its target cell. These are individual restrictions; the remaining collision checks still couple different factors.

By Proposition 7.4, the primary marginal of each path factor has density at most G relative to P_i , and integrates to one. For the extra point the density is $\mathbf{1}_{A_k} / |A_k|$ relative to P_* , which is at most $1/0.98$. Enlarge the fixed G to be at least two, covering this factor as well. These marginal densities and their auxiliary kernels depend on the exposed data and the route and label choices; they do not depend on the unselected unexposed paths.

The union bound proving Equation (9.6) verifies hypothesis (i) for each proposal factor. The same argument with P_i in place of $\Pi_i^{A_k}$ verifies it for insertion of a selected reference path, by Proposition 7.4. The reference extra point has no checks. These estimates are uniform over all individually permitted opponent paths, whether or not their labels are accessible, and deleting checks can only improve them. Thus the bound remains available for every reduced normalizer in Lemma 9.3.

Take two independent route and label choices at the same data and endpoint pair, and denote their selected index sets by S, T . For the chosen primary side, declare a factor of S near if its center is at lattice supremum distance at most R from some center of T , and declare every other factor far. Define C_S, F_S, C_T, F_T accordingly. A shared real label has centers at distance at most two in the two assignments, so it is near; the shared extra index has the same endpoint center. Consequently the far index sets are disjoint from the opposing lists.

All paths carried by a factor lie within distance eleven of its center. The original path of an opposing real label also lies within distance eleven of that opposing factor's center. Since $R = 30$, far factors cannot interact with any coordinate carried by the opposite list or with those original paths. This includes the auxiliary-copy paths. Far sets therefore meet the separation assumptions of Lemma 9.3.

For clarity, a reduced far proposal retains exactly the paths carried by its far vertex factors and the fixed paths outside $S \cup T$. It deletes the omitted near factors in both copies. It does not put their original trajectories back as unselected obstacles. With this convention all the normalizers in (9.11)–(9.14) are the ones arising from the present construction.

Write L_S, L_T for the two primary likelihoods. If

$$J_R = \#\{(j, k) : d_\infty(z_j, z'_k) \leq R\},$$

then $|C_S| + |C_T| \leq 2J_R$. Lemma 9.3 gives, at the fixed data,

$$\int L_S L_T d(\lambda_{\mathcal{I}} \otimes P_*) \leq G^{|S \cap T|} (1 - \chi_D)^{-4J_R}. \quad (9.15)$$

In particular, no factor is paid for a long portion of one route that stays away from the other.

9.6 Averaging labels and forgetting the paths

Condition on the data and the two lattice paths before choosing their labels. Each common departure cell has at least $M/2$ accessible labels. The two uniform choices at that cell match with probability at most $2/M$, and the match events are independent between distinct common cells. Real labels anchored at different cells cannot agree. The shared extra point contributes one factor G , so

$$\mathbb{E}_{\text{labels}} G^{|S \cap T|} \leq G \exp\left(\frac{2(G-1)}{M} J_R\right). \quad (9.16)$$

Choose the fixed integer M sufficiently large for the good-site supply and for

$$\frac{2(G-1)}{M} \leq \frac{\kappa}{2}. \quad (9.17)$$

Then increase D_* so that, for every $D \geq D_*$ in the allowed α window, all earlier estimates hold and

$$-4 \log(1 - \chi_D) \leq \frac{\kappa}{2}, \quad \frac{G_D}{D} \geq \frac{\ell}{8M}. \quad (9.18)$$

The first requirement is possible by (9.6); the second follows once $\ell D/(4M) \geq 2$. These choices do not change G , κ , or C_{path} .

The following use of an averaged likelihood and a two-route second moment also occurs in the point-reassignment argument of Peres and Sly [38, Proposition 2.1]. Here Lemma 9.3 supplies the additional conditional-normalizer estimate required by the hard-core path law.

Let $\bar{L}$ be the likelihood averaged over the conditional route and label choices, with value zero off $\mathcal{E}_{vw}$. Tonelli's theorem expresses its square integral using two independent such choices. Combining (9.15)–(9.18) and the annealed encounter bound in Proposition 8.1 gives

$$\mathbb{E}_{\mathcal{B}} \int \bar{L}^2 d(\lambda_{\mathcal{I}} \otimes P_*) \leq G \mathbb{E}[\mathbf{1}_{\mathcal{E}_{vw}} e^{\kappa J_R}] \leq C_2, \quad C_2 := G C_{\text{path}}. \quad (9.19)$$

Here the two route randomizations share the same data; the expectation over the data is taken only once. The event is not conditioned to have probability one. Its indicator is part of both likelihoods,

and its square is itself. The calculation applies separately with either copy primary, with the same constant C_2 .

We next check carefully that tagging and observing the bath do not increase this bound. At fixed data, split $\bar{L} = \sum_{i \in \mathcal{I}} f_i$, where f_i contains exactly the choices designating i as the primary tag. The functions are nonnegative. In the first copy f_i is supported where $\omega_i(t) \in B_v$; in the second it is supported where $\omega_i(t) \in B_w$. Since

$$\sum_i f_i^2 \leq \left(\sum_i f_i \right)^2,$$

the measure that retains the tag index has squared likelihood integral at most C_2 relative to the *sum* of the tag-restricted reference measures on the disjoint copies indexed by i . Now apply the observation kernel that orders the bath and retains $\mathcal{B}, x, y, X'$, and forget the tag and all the paths. If a finite measure has density f relative to a reference P , the density after applying a common kernel is the conditional average of f in the corresponding joint reference measure. Conditional Cauchy–Schwarz bounds its squared integral by $\int f^2 dP$. This remains valid for finite reference measures, by normalization when their mass is nonzero.

For either primary copy, the resulting measure is the common observed $\hat{\nu}$ already constructed. Its corresponding reference is precisely $\hat{P}_g$ or $\hat{Q}_g$. We have proved

$$\int \left(\frac{d\hat{\nu}}{d\hat{P}_g} \right)^2 d\hat{P}_g \leq C_2, \quad \int \left(\frac{d\hat{\nu}}{d\hat{Q}_g} \right)^2 d\hat{Q}_g \leq C_2. \quad (9.20)$$

Together with (9.7), Lemma 9.1 now gives

$$\text{Aff}(\hat{P}_g, \hat{Q}_g) \geq \frac{1}{4C_2}. \quad (9.21)$$

This lower bound is uniform in the group index and in every admissible ordered pair of cells.

Proof of Theorem 2.1. With the fixed choices above, set

$$c_* := \frac{v\ell}{32MGC_{\text{path}}} > 0, \quad v = \frac{1}{2000}. \quad (9.22)$$

For large K , at least vV^2 ordered cell pairs meet the displacement condition of Proposition 8.1. For each such pair, Lemma 9.2, (9.18) and (9.21) give

$$\text{Aff}(P_{vw}, Q_{vw}) \geq \frac{G_D}{4DC_2} \geq \frac{\ell}{32MC_2}.$$

Sum these nonnegative contributions in (9.3) to obtain $B_{\text{occ}} \geq v\ell/(32MC_2)$, which is (9.22). Every estimate used above is uniform over the stated temperature interval. The volume threshold is locally uniform for D in compact subintervals of $[D_*, \infty)$, by the slab, bridge and group estimates and Proposition 8.1. No parameter entering c_* is chosen after D_* , so increasing D_* does not change the occupation bound.

For completeness, the choices leading to this bound have no circular dependence. Fix the lattice radius $R = 30$ and its bad-site tolerance from Proposition 8.1. Make the spatial choices of large α_0 , then ℓ, θ and small b as in Sections 3 and 4. Choose a sufficiently small target certificate-failure base p_1 . Proposition 5.4 then allows H_1 to be fixed before choosing a sufficiently small positive t , also meeting the deterministic bridge requirements. This fixes the density bound G . Choose M

large enough for the group supply and (9.17); finally choose D_* large enough for all small-scale requirements and (9.18). The required lower bound on K is imposed afterwards. In particular $\beta \geq D$ is the only temperature restriction; no condition relates β to the volume. $\square$

The remaining step is the return to fixed physical density and temperature.

10 Return to fixed physical parameters

We now prove Theorem 1.1 from the scaled estimate. This step also makes explicit why the temperature remains positive throughout the thermodynamic limit.

Let α_0, D_*, c_* be furnished by Theorem 2.1. For a fixed exclusion distance $a > 0$, set

$$\rho_*(a) = \frac{\alpha_0^3}{4D_*^2 a^3}.$$

Fix $0 < \rho < \rho_*(a)$ and define

$$s_\infty = \sqrt{\frac{\alpha_0}{a\rho}}, \quad D_\infty = \frac{\alpha_0^{3/2}}{\sqrt{\rho a^3}}, \quad T_{\text{phys}} = \frac{1}{4s_\infty^2 D_\infty} > 0. \quad (10.1)$$

These are fixed numbers depending on a, ρ , and the constants of the proof. In particular, $D_\infty > 2D_*$.

Consider any sequence $L \rightarrow \infty$, $N/L^3 \rightarrow \rho$. Choose

$$K = \left\lfloor \sqrt{\frac{aN}{\alpha_0 L}} \right\rfloor, \quad s = L/K, \quad D = N/K^3, \quad r = a/s.$$

Then $K \rightarrow \infty$ and

$$s \rightarrow s_\infty, \quad D \rightarrow D_\infty, \quad Dr = \frac{aN}{LK^2} \rightarrow \alpha_0.$$

Thus the scaled core has the required form $r = \alpha/D$ with $\alpha \in [\alpha_0/2, 2\alpha_0]$ eventually. Under the unitary dilation $x = s\tilde{x}$, the Hamiltonian becomes $s^{-2}H_N$ on Λ_K , and the canonical Gibbs state has scaled temperature $T = s^2 T_{\text{phys}}$. Since

$$DT = Ds^2 T_{\text{phys}} \rightarrow \frac{1}{4},$$

we have $0 < T \leq D^{-1}$ eventually. The local uniformity of the volume threshold in Theorem 2.1 applies because D remains in a compact neighborhood of D_∞ .

The dilation sends $u_{0,L}$ to u_0 and preserves the occupation fraction. Hence Theorem 2.1 gives, eventually along every such sequence,

$$\frac{\langle u_{0,L}, \gamma_{N,L,T_{\text{phys}}}^{(1)} u_{0,L} \rangle}{N} \geq c_*.$$

Taking the lower limit proves Theorem 1.1. $\square$

References

- [1] Emmanuel Abbe, Laurent Massoulié, Andrea Montanari, Allan Sly, and Nikhil Srivastava. Group synchronization on grids. *Mathematical Statistics and Learning*, 1(3/4):227–256, 2018. doi: 10.4171/MSL/6.
- [2] Michael Aizenman, Elliott H. Lieb, Robert Seiringer, Jan Philip Solovej, and Jakob Yngvason. Bose-Einstein quantum phase transition in an optical lattice model. *Physical Review A*, 70(2):023612, August 2004. doi: 10.1103/PhysRevA.70.023612.
- [3] Ery Arias-Castro, Emmanuel J. Candès, Hannes Helgason, and Ofer Zeitouni. Searching for a trail of evidence in a maze. *The Annals of Statistics*, 36(4):1726–1757, 2008. doi: 10.1214/07-AOS526. URL <https://doi.org/10.1214/07-AOS526>.
- [4] Giulia Basti, Morris Brooks, Serena Cenatiempo, Alessandro Olgiati, and Benjamin Schlein. Lee–Huang–Yang energy for dilute Bose gases: an upper bound for general potentials, 2026. URL <https://arxiv.org/abs/2609.25314v1>. Version 1, September 21, 2026.
- [5] Giulia Basti, Morris Brooks, Serena Cenatiempo, Alessandro Olgiati, and Benjamin Schlein. The Lee–Huang–Yang energy for a dilute gas of hard spheres: an upper bound, 2026. URL <https://arxiv.org/abs/2603.13084v2>. Version 2, September 21, 2026.
- [6] Itai Benjamini, Robin Pemantle, and Yuval Peres. Unpredictable paths and percolation. *The Annals of Probability*, 26(3):1198–1211, 1998. doi: 10.1214/aop/1022855749. URL <https://arxiv.org/abs/math/9701227>.
- [7] Noam Berger and Yuval Peres. Detecting the trail of a random walker in a random scenery. *Electronic Journal of Probability*, 18(87):1–18, 2013. doi: 10.1214/EJP.v18-2367.
- [8] N. N. Bogoliubov. On the theory of superfluidity. *Journal of Physics (USSR)*, 11(1):23–32, 1947. URL https://ufn.ru/pdf/jphysussr/1947/11_1/3jphysussr19471101.pdf.
- [9] Satyendra Nath Bose. Plancks Gesetz und Lichtquantenhypothese. *Zeitschrift für Physik*, 26(1):178–181, 1924. doi: 10.1007/BF01327326. URL <https://doi.org/10.1007/BF01327326>.
- [10] Jacky J. Chong, Hao Liang, and Phan Thành Nam. Kinetic localization via Poincaré-type inequalities and applications to the condensation of Bose gases. *Journal of Functional Analysis*, 291(5):111542, 2026. doi: 10.1016/j.jfa.2026.111542. URL <https://www.sciencedirect.com/science/article/abs/pii/S0022123626002065>.
- [11] Jan Dereziński and Christian Gérard. Asymptotic completeness in quantum field theory: Massive Pauli–Fierz hamiltonians. *Reviews in Mathematical Physics*, 11(4):383–450, 1999. doi: 10.1142/S0129055X99000155.
- [12] Andreas Deuchert and Robert Seiringer. Gross–Pitaevskii limit of a homogeneous Bose gas at positive temperature. *Archive for Rational Mechanics and Analysis*, 236(3):1217–1271, 2020. doi: 10.1007/s00205-020-01489-4.
- [13] Andreas Deuchert and Robert Seiringer. Semiclassical approximation and critical temperature shift for weakly interacting trapped bosons. *Journal of Functional Analysis*, 281(6):109096, 2021. doi: 10.1016/j.jfa.2021.109096.

- [14] Andreas Deuchert, Robert Seiringer, and Jakob Yngvason. Bose–Einstein condensation in a dilute, trapped gas at positive temperature. *Communications in Mathematical Physics*, 368(2): 723–776, 2019. doi: 10.1007/s00220-018-3239-0. URL <https://d-nb.info/1169529054/34>.
- [15] Freeman J. Dyson. Ground-state energy of a hard-sphere gas. *Physical Review*, 106:20–26, 1957. doi: 10.1103/PhysRev.106.20.
- [16] Albert Einstein. Quantentheorie des einatomigen idealen Gases. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, pages 261–267, 1924. URL <https://echo-old.mpiwg-berlin.mpg.de/ECHODocuView?url=%2Fpermanent%2Fecho%2Feinstein%2Fsitzungsberichte%2FPG8B073X%2Findex.meta&viewMode=index>.
- [17] Albert Einstein. Quantentheorie des einatomigen idealen Gases. Zweite Abhandlung. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, pages 3–14, 1925. URL <https://echo-old.mpiwg-berlin.mpg.de/ECHODocuView?url=%2Fpermanent%2Fecho%2Feinstein%2Fsitzungsberichte%2FR1Y4X6GP%2Findex.meta&viewMode=index>.
- [18] Richard P. Feynman. Atomic theory of the λ transition in helium. *Physical Review*, 91(6):1291–1301, 1953. doi: 10.1103/PhysRev.91.1291. URL <https://doi.org/10.1103/PhysRev.91.1291>.
- [19] S. Fournais, L. Junge, T. Girardot, L. Morin, M. Olivieri, and A. Triay. The free energy of dilute Bose gases at low temperatures interacting via strong potentials. *Annales Henri Poincaré*, 2026. doi: 10.1007/s00023-026-01676-1. URL <https://link.springer.com/article/10.1007/s00023-026-01676-1>. Published online 9 March 2026.
- [20] Søren Fournais. Length scales for BEC in the dilute Bose gas. In Pavel Exner, Rupert L. Frank, Fritz Gesztesy, Helge Holden, and Timo Weidl, editors, *Partial Differential Equations, Spectral Theory, and Mathematical Physics: The Ari Laptev Anniversary Volume*, volume 18 of *EMS Series of Congress Reports*, pages 115–133. EMS Press, 2021. doi: 10.4171/ECR/18-1/7. URL <https://ems.press/books/ecr/187/3520>.
- [21] Søren Fournais and Jan Philip Solovej. The energy of dilute Bose gases II: the general case. *Inventiones Mathematicae*, 232(2):863–994, 2023. doi: 10.1007/s00222-022-01175-0.
- [22] Stefano Galanda and Nicola Pinamonti. Equilibrium states for non relativistic Bose gases with condensation, 2025. URL <https://arxiv.org/abs/2509.25101>.
- [23] Stefano Galanda and Nicola Pinamonti. Equilibrium states for non relativistic Bose gases and the Gross-Pitaevskii limit, 2026. URL <https://arxiv.org/abs/2606.25775>.
- [24] Christophe Garban and Thomas Spencer. Continuous symmetry breaking along the Nishimori line. *Journal of Mathematical Physics*, 63(9):093302, 2022. doi: 10.1063/5.0087024. URL <https://arxiv.org/abs/2109.01617v3>. Theorem locators refer to arXiv version 3, 27 March 2023.
- [25] Olle Häggström and Elchanan Mossel. Nearest-neighbor walks with low predictability profile and percolation in $2 + \varepsilon$ dimensions. *The Annals of Probability*, 26(3):1212–1231, 1998. doi: 10.1214/aop/1022855750. URL https://faculty.wharton.upenn.edu/wp-content/uploads/2014/07/Nearest_Neighbor_Walks_with_Low_Predictability_Profile_and_Percolation_1.pdf.
- [26] Lukas Junge. Propagation of condensation via Neumann localization in the dilute Bose gas, 2026. URL <https://arxiv.org/abs/2603.20776v3>. Version 3, 4 April 2026.

- [27] T. D. Lee, Kerson Huang, and C. N. Yang. Eigenvalues and eigenfunctions of a Bose system of hard spheres and its low-temperature properties. *Physical Review*, 106(6):1135–1145, June 1957. doi: 10.1103/PhysRev.106.1135.
- [28] Mathieu Lewin. Geometric methods for nonlinear many-body quantum systems. *Journal of Functional Analysis*, 260:3535–3595, 2011. doi: 10.1016/j.jfa.2010.11.017.
- [29] Elliott H. Lieb and Mary Beth Ruskai. Proof of the strong subadditivity of quantum-mechanical entropy. *Journal of Mathematical Physics*, 14(12):1938–1941, 1973. doi: 10.1063/1.1666274.
- [30] Elliott H. Lieb and Robert Seiringer. Proof of Bose–Einstein condensation for dilute trapped gases. *Physical Review Letters*, 88:170409, 2002. doi: 10.1103/PhysRevLett.88.170409. URL <https://arxiv.org/abs/math-ph/0112032>.
- [31] Elliott H. Lieb and Jakob Yngvason. Ground state energy of the low density Bose gas. *Physical Review Letters*, 80:2504–2507, 1998. doi: 10.1103/PhysRevLett.80.2504. URL <https://arxiv.org/abs/cond-mat/9712138>.
- [32] Elliott H. Lieb, Robert Seiringer, and Jakob Yngvason. Bosons in a trap: A rigorous derivation of the Gross–Pitaevskii energy functional. *Physical Review A*, 61:043602, 2000. doi: 10.1103/PhysRevA.61.043602.
- [33] Göran Lindblad. Completely positive maps and entropy inequalities. *Communications in Mathematical Physics*, 40(2):147–151, 1975. doi: 10.1007/BF01609396.
- [34] Douglas Lundholm, Fabian Portmann, and Jan Philip Solovej. Lieb–Thirring bounds for interacting Bose gases. *Communications in Mathematical Physics*, 335:1019–1056, 2015. doi: 10.1007/s00220-014-2278-4.
- [35] OpenAI. A density-uniform condensate bound for dilute Bose gases. OpenAI Math Release preprint [OAI:A-density-uniform-condensate-bound-for-dilute-Bose-gases-September-27-2026](#), 2026.
- [36] OpenAI. Ground-state condensation in the dilute hard-sphere gas. OpenAI Math Release preprint [OAI:Ground-state-condensation-in-the-dilute-hard-sphere-gas-September-24-2026](#), 2026.
- [37] Oliver Penrose and Lars Onsager. Bose–Einstein condensation and liquid helium. *Physical Review*, 104(3):576–584, 1956. doi: 10.1103/PhysRev.104.576.
- [38] Yuval Peres and Allan Sly. Rigidity and tolerance for perturbed lattices, 2014. URL <https://arxiv.org/abs/1409.4490>.
- [39] András Sütő. Bose–Einstein condensation of interacting bosons: A two-step proof, 2023. URL <https://arxiv.org/abs/2305.18959>. Version 1, May 30, 2023.
- [40] Ádám Timár. Boundary-connectivity via graph theory. *Proceedings of the American Mathematical Society*, 141(2):475–480, 2013. doi: 10.1090/S0002-9939-2012-11333-4. URL <https://arxiv.org/abs/0711.1713v2>.
- [41] Daniel Ueltschi. Feynman cycles in the Bose gas. *Journal of Mathematical Physics*, 47:123303, 2006. doi: 10.1063/1.2383008. URL <https://arxiv.org/abs/math-ph/0605002>.